When the Detector Is Wrong: A Controlled Study of Post-Training Backdoor Removal Under Imperfect Recall

Thota Venkata Sai, Sarang N S, Divyansh Tiwari, and Urja Mandali

https://github.com/venkatasai0852/fml-backdoor-recall-cliff

***Abstract*—Post-training removal of backdoors is normally evaluated with an oracle forget set, yet a practitioner must act on detector output that is neither complete nor precise. We present a controlled study of how detector recall and precision govern removal success. Forget sets $\mathscr{F}(r,p)$ are constructed to contain a specified fraction of the poison at a specified precision, and five removal families are swept over ten (r,p) cells and three seeds, with the critical cells repeated on three further architecture–attack combinations. Deletion-based removal is a step function of the poison left behind: retraining on the retained set meets the success criterion only at r=1.0, because at r=0.95 attack success is still low but the clean-accuracy cost already breaks the criterion, and at r=0.90 the backdoor returns at full strength. A repair-and-inoculate recipe tolerates r=0.5, but only because it uses the known trigger; when the trigger is synthesised instead, the overlap with the true patch is zero and attack success stays above 95%. Two proposed refinements fail at scale: calibrate-then-remove cannot find a usable proxy threshold, and partial unlearning does not amplify the detectability of missed poison. Apparent removal by feature-space ascent is a batch-norm artefact. Durability probes on the oracle-deleted model show suppression rather than erasure: one hundred poisoned samples restore 97.7% attack success within ten steps. On RadioML automatic modulation classification the same recall cliff appears in the minimum recall γ^* as a function of SNR. We conclude that post-training removal is bounded by detector recall, and that trigger-aware repair is a ceiling for the field rather than a deployable blind defence.**

***Index Terms*—Backdoor defence, data poisoning, machine unlearning, detector recall, trustworthy machine learning, radio modulation classification.**

I. Introduction

A backdoor is implanted by training on a dataset in which a small fraction of inputs carry a trigger and a target label. The resulting network classifies clean inputs normally but maps any trigger-stamped input to the attacker's class, often with an attack success rate (ASR) above 99% while clean accuracy moves by less than a point. Because the poison fraction can be as low as 1%, inspection of the dataset is unreliable, and the standard remedy is a two-stage pipeline: a detector ranks or flags training samples, and a removal method then acts on the flagged set, either by deleting it and retraining or by post-training unlearning.

The second stage is usually studied in isolation. Published evaluations of removal methods typically hand the method the true poison set, or a superset that contains it, and report how much attack success falls and how much clean accuracy is retained. This is a sensible way to compare algorithms, but it answers a question that a practitioner cannot ask. In deployment the forget set comes from a detector, and detectors are neither complete nor precise. On our own measurements, activation clustering at a 5% poison rate flags 4,140 samples of which 2,500 are poison (precision 0.60, recall 1.00), spectral signatures flags 3,750 with precision 0.64 and recall 0.96, and at a 10% poison rate activation clustering returns no flags at all. A removal method that assumes a clean forget set is therefore being evaluated outside the regime in which it would be used.

This paper treats the composition of the forget set as the experimental variable (Fig. 1). We do not propose a new detector and we do not assume that any particular detector is the right one. Instead we construct forget sets $\mathscr{F}(r,p)$ that contain a controlled fraction of the poison at a controlled precision, and we ask how the recall r and precision p of that set govern the outcome of five removal families. The construction is deterministic given (r,p) and the seed, it does not depend on a detector's score distribution, and it lets us place the boundary of success precisely rather than anecdotally. We sweep ten (r,p) cells and three seeds on one architecture–attack combination, repeat the critical cells on three further combinations, and add two studies that the grid cannot answer: whether a removed model stays removed under relearning, and whether the same boundary appears in a different modality. For the second we use RadioML automatic modulation classification, where the trigger is a synthetic in-phase/quadrature pattern and the defence metric becomes a function of signal-to-noise ratio (SNR).

Our results are mostly negative, and we report them as such. The main findings are the following.

1) **Deletion is a cliff, not a slope.** Retraining on the retained set meets both halves of the success criterion only when the forget set contains every poisoned sample. At r=0.95 ASR is still low (2.84 ± 0.73%), but clean accuracy falls to 84.58%, four points below the gold model, so the joint criterion already fails; at r=0.90 the backdoor returns at full strength (ASR 100.00 ± 0.00%), and at r=0.25 it is equally alive. The smallest recall at which deletion succeeds on all three seeds is therefore r^*=1.00.
2) **Recall dominates precision.** At fixed r=0.90, lowering precision from 1.0 to 0.7 adds 964 clean samples to the

The authors are with the Department of Electronics and Communication Engineering, Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat 395007, India.

forget set (3,214 against 2,250) and changes deletion's ASR from 100.00% to 99.99%. Missing poison is categorically worse than deleting extra clean data.

3) **Repair tolerates half the poison, but only with the trigger.** A recipe that relabels flagged samples by class centroid and then inoculates the model against the trigger holds ASR at or below 3.6% down to r=0.5, with a clean-accuracy cost of 0.7–1.3 points. The tolerance is real but bounded by knowledge: the inoculation step uses the true trigger. When the trigger is synthesised from the model instead, its intersection with the true patch is empty (IoU = 0.000) and ASR remains at 98.99% (BadNets) and 95.72% (Blend). The recipe is an upper bound on what trigger-aware repair can do, not a blind defence.
4) **Two proposed refinements fail.** Calibrate-then-Remove never finds a proxy-removed model whose trigger response falls below 5% (the minimum over twenty thresholds is 97.5%), so it degenerates to its fallback and leaves ASR at 99.97%. Amplify-and-detect, the hypothesis that partial unlearning makes missed poison easier to detect, does not survive at scale: over three seeds and a 10% poison rate, the area under the ROC curve (AUC) separating missed poison from clean data stays between 0.468 and 0.503, against a pre-registered promotion gate of base +0.10.
5) **Apparent unlearning can be a normalisation artefact.** Feature-space ascent appears to remove the backdoor (ASR 0.18% at learning rate 10^{-3}) only by destroying the network; repeating the identical schedule with batch-norm statistics frozen to the retain set leaves ASR at 99.94–99.98% and clean accuracy intact.
6) **Removal suppresses; it does not erase.** Even the oracle-deleted gold model returns to 97.7% ASR within ten gradient steps on one hundred fresh poisoned samples, its membership-inference AUC sits at chance, and the clean-versus-trigger KL gap characteristic of a live backdoor has vanished. The backdoor is dormant, not gone.
7) **The cliff is not specific to images.** On RadioML 2016.10a the same step appears: deletion needs full recall, while trigger-aware repair holds at r=0.5 for SNR at or above 0 dB. At lower SNR the classifier is near chance and the ASR criterion is vacuous; we mark that region explicitly rather than reporting a spurious tolerance.

The paper is organised as follows. Section II reviews backdoor attacks, detection, and removal. Section III formalises the forget set, the success criterion, and the tolerance r^*. Section IV describes the removal methods and the two detectors. Section V fixes the experimental protocol for CIFAR-10 and RadioML. Sections VI–VIII report the recall cliff, its transfer across architectures and attacks, and the two failed refinements. Section IX reports durability and the efficiency-oriented baselines, Section X the radio study, and Sections XI–XIII discuss implications, limitations, and conclude. Every number in every table traces to a per-run CSV record produced by the frozen drivers described in Section V; no value was transcribed by hand. The per-run records, the drivers, and the asset builder are available in the accompanying repository described in Section XIII.

II. Related Work

A. Backdoor Attacks

Backdoor poisoning was established by Gu et al. with BadNets, a small checkerboard patch stamped onto a fraction of training images and relabelled to a target class [1]. Liu et al. showed that the same effect can be obtained by perturbing weights rather than data [2], and Chen et al. gave an early data-poisoning formulation with a learned trigger [3]. Later work made the trigger stealthier: clean-label attacks that require no label change [4], hidden-trigger attacks that optimise the poisoned input to be far from the trigger at test time [5], warping-based triggers [6], learnable and imperceptible triggers [7], input-aware triggers [8], and physically realisable triggers [9]. In federated learning, a single malicious client can inject a backdoor that survives aggregation [10], [11], [64]. Surveys by Li et al. [12] and Goldblum et al. [13] organise this space and stress that defence evaluation is fragmented.

Backdoor poisoning is a special case of data poisoning, which has a longer history: support-vector poisoning [60], certified defences against poisoning [61], poisoning of regression learning [62], and attacks that break data sanitization at scale [44], including gradient-matching poisons that transfer across models [63]. The lesson of that line of work, that a defence tuned to one poisoning strength can fail at another, motivates the sweep over recall and precision here.

The design space has since widened well beyond dirty-label patches: clean-label poisoning of feature extractors [43], subpopulation poisoning [45], frequency-domain triggers [48], reflection triggers [46], dynamic triggers [47], and latent backdoors [49]; contrastive encoders [52]; and instruction-tuned language models [53]. Data-sanitization defences can themselves be evaded by stronger poisoning [44], and BackdoorBench consolidates implementations and evaluation protocols across this space [51].

Our threat model is the simplest and most studied one: an attacker controls a fraction of a static training set and stamps a fixed, visible trigger. We use two instances, a 4×4 checkerboard and a blended pattern, precisely because they are canonical. The contribution of this paper is not a new attack but the measurement of what happens after one is detected imperfectly.

B. Detection

Detection methods score inputs, features, or model parameters. Neural Cleanse optimises a minimal trigger per class and flags a class whose trigger is anomalously small [14]. ABS stimulates neurons and observes whether a single output class dominates [15]. STRIP perturbs inputs with superimposed patterns and measures prediction entropy [16]. Activation clustering projects per-class activations and separates them into two clusters [17], and spectral signatures analyses the

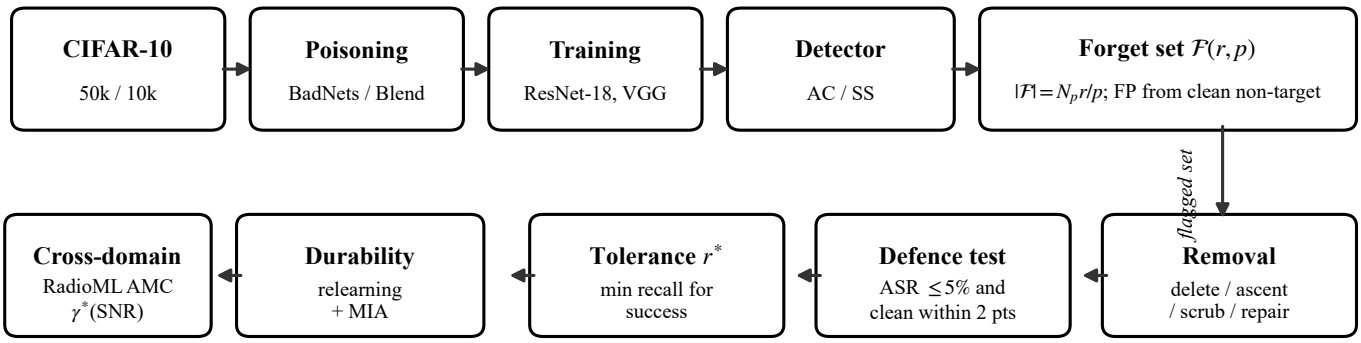


Fig. 1. The measurement pipeline. A poisoned training set yields a backdoored model, a detector flags a subset, and the flagged set is converted into a forget set $\mathcal{F}(r,p)$ with controlled recall and precision. Removal is then scored on attack success and clean accuracy, summarised as the tolerance r^*, probed for durability under relearning, and repeated on a radio-modulation task.

top singular direction of per-class representations [18]. Fine-pruning observes that backdoored neurons are dormant on clean inputs and prunes them [19]. Post-training scanners continue to broaden coverage; MM-BD, for example, scores classes by a maximum-margin statistic and does not assume a patch-shaped trigger [50].

All of these methods trade missed detections against false positives, and their operating points depend on poison rate, architecture, and attack. Detection is therefore not a preprocessing detail that can be assumed away: it defines the input to the removal stage. Recent defences continue to improve the detector side, including variational formulations that jointly model trigger and class-conditional structure [20]. Our study is complementary: we take detector output as given and ask how much of its error the removal stage can absorb. To keep the manipulation interpretable we parameterise that output by recall and precision rather than by a detector-specific threshold, and we anchor the parameterisation to two real detectors in Section VI.

C. Removal and Unlearning

The gold standard for removal is retraining on the retained data [1], [19]. Because full retraining is expensive, a large literature approximates it. SISA shards the training set, trains a constituent model per shard, and unlearns by retraining only the affected shards [21]. Amnesiac unlearns by ascending the loss on the forget set [22], Eternal Sunshine approximates the retrained weights in a quadratic Fisher sense [23], and certified removal provides a differential-privacy-style guarantee for convex or strongly convex models [24]. SCRUB trains a student to match the teacher on the retain set while disagreeing on the forget set [25]. Influence functions and gradient tracing estimate which training points mattered and can be used to down-weight them [26], [27], and example-forgetting scores have been proposed as a way to choose which points a model should forget [70]. Suriyakumar and Wilson characterise the limits of approximate-removal guarantees [28]. Surveys of machine unlearning [55] and domain-specific benchmarks [56], [57] have grown around this literature, and what an unlearning claim should certify is now a central question for language models [59].

A parallel family acts during or immediately after training rather than on a forget set: anti-backdoor learning separates poison from clean data while training [65], decoupled training withholds the trigger association from the feature extractor [66], adversarial neuron pruning removes trigger-sensitive units [67], and attention distillation fine-tunes the backdoored model on clean data [68]. Trigger-reverse-engineering scanners reconstruct a candidate trigger before removal [69]. These methods reduce the need for an exact forget set, but each still needs either clean data, a clean reference model, or a reconstructed trigger; none is evaluated under a forget set whose recall is known to be below one.

Closest to our recipe are methods that edit the association rather than delete the data. Zhao and Wressnegger study the symmetry between learning a backdoor and removing one [29]. Repair-based pipelines relabel flagged samples and fine-tune, optionally with an inoculation step that presents the trigger with its correct label. These methods are attractive under imperfect detection because they do not require the forget set to be exact, but their inoculation step needs the trigger. We quantify both sides of that trade: the tolerance the recipe buys, and the knowledge it consumes.

Two 2025 results shape our interpretation. Pawelczyk et al. show that approximate unlearning methods, including methods designed for large language models, fail to remove the effect of poisoned data even when they appear to succeed on standard unlearning metrics [30]. Hu et al. show that benign relearning restores supposedly unlearned behaviour [31], and the same persistence under benign retraining has been observed in language models trained to exhibit deceptive behaviour [54]. Our durability study is a vision-domain analogue of the second result, and our overall conclusion is consistent with the first: unlearning metrics that do not probe the attack after removal can be misleading.

D. Evaluation Under Imperfect Detection

The gap we address is evaluative. Removal papers usually construct the forget set from ground-truth poison labels, or

from a detector threshold that is tuned until recall is perfect. Detector papers report recall and precision but stop before removal. Studies that connect the two usually vary a single threshold and report a single operating point, which makes it hard to separate the effect of missed poison from the effect of deleted clean data. Controlled manipulation of both quantities, across method families and across domains, is what our grid provides. The tolerance r^* is intended as a reusable summary: it answers "how good must the detector be for this removal method to work?" in the same units for every method.

E. Calibration and Risk Control

Calibrate-then-Remove, one of the two refinements we test, borrows the logic of distribution-free risk control. Conformal prediction converts a score into a prediction set with a user-specified coverage guarantee under exchangeability [32], [33], and conformal risk control extends the guarantee from coverage to an arbitrary bounded loss [58]. The same machinery can in principle choose a removal threshold that controls the risk of a live backdoor. The guarantee depends on a proxy that is exchangeable with the test trigger. Our experiment tests whether a synthesised trigger can serve as that proxy; the answer, on both attacks, is no.

F. Positioning

This paper is a measurement study with one attempted mechanism and an explicit negative result. It does not claim a new attack, a new detector, or a deployable defence. Its contributions are the controlled forget-set construction, the recall cliff and its transfer across architectures and attacks, the tolerance metric r^*, the demonstration that trigger synthesis breaks the only method that tolerates imperfect recall, and the replication of the cliff in a radio modulation task. The repair recipe we study is a strong baseline and an upper bound; we label it as such wherever it appears.

III. Problem Formulation and Metrics

A. Setup

Let $\mathscr{D} = \{(x_i, y_i)\}_{i=1}^{N}$ be a training set and let $\mathscr{P} \subset \mathscr{D}$ be the poisoned subset, $|\mathscr{P}| = N_p$. A model $f = \mathscr{A}(\mathscr{D})$ is produced by a fixed training algorithm $\mathscr{A}$; in this paper $\mathscr{A}$ is the schedule of Table I. For a target class y_t and a trigger function T, the attack success rate is

$$\mathrm{ASR}(f) = \frac{1}{|\mathscr{D}_{\text{test}}|} \sum_{(x,y)\in\mathscr{D}_{\text{test}}} \mathbb{1}[f(T(x)) = y_t], \tag{1}$$

and clean accuracy is the usual top-1 accuracy on unmodified test inputs. A backdoor is considered present when ASR is high and clean accuracy is close to that of a model trained without poison. The gold reference is $f_{\text{gold}} = \mathscr{A}(\mathscr{D} \setminus \mathscr{P})$, trained on the same data with the true poison removed. All success statements are relative to f_{gold}, not to the poisoned model, because the poisoned model's clean accuracy can be slightly higher or lower by chance and would otherwise make weak methods look adequate.

Algorithm 1 Controlled forget-set construction

Require: training set $\mathscr{D}$, poison mask, r, p, seed s
Ensure: $\mathscr{F}(r,p)$ with $|\mathscr{F}| = N_p r/p$
$\mathscr{P} \leftarrow$ poisoned indices; $N_p \leftarrow |\mathscr{P}|$
$n_A \leftarrow \mathrm{round}(N_p r)$
$n_B \leftarrow \mathrm{round}(N_p r(1-p)/p)$
$A \leftarrow$ sample n_A indices from $\mathscr{P}$ with seed s
$C \leftarrow$ clean, non-target training indices
$B \leftarrow$ sample n_B indices from C with seed s
return $A \cup B$

B. Detectors and Their Operating Points

A detector is a map $\mathrm{DET} : \mathscr{D} \to 2^{\mathscr{D}}$ producing a flagged set S. Relative to the ground-truth poison,

$$r = \frac{|S \cap \mathscr{P}|}{|\mathscr{P}|}, \qquad p = \frac{|S \cap \mathscr{P}|}{|S|}, \tag{2}$$

with p defined as zero when $S = \emptyset$. Recall r is the fraction of poison the removal method will see; precision p is the fraction of the flagged set that is genuinely poisoned. Both matter, but they enter the removal problem asymmetrically: a missed poison sample remains in the training set and can sustain the backdoor, whereas a falsely flagged clean sample is merely removed from training. The experiments are designed to expose that asymmetry.

C. Controlled Forget Sets

Rather than threshold a detector score, we construct the forget set directly. For requested recall $r \in [0,1]$ and precision $p \in (0,1]$, let

$$\mathscr{F}(r,p) = A \cup B, \quad A \subseteq \mathscr{P}, \ |A| = \mathrm{round}(N_p r), \tag{3}$$

$$B \subseteq \mathscr{D}_{\text{clean}} \setminus \mathscr{C}_{y_t}, \quad |B| = \mathrm{round}\left(\frac{N_p r(1-p)}{p}\right), \tag{4}$$

where $\mathscr{C}_{y_t}$ is the set of clean training samples of the target class. The sizes follow from the definition of precision: if $|A| = N_p r$ and $|A|/|A \cup B| = p$, then $|\mathscr{F}| = N_p r/p$ and $|B| = N_p r(1-p)/p$. False positives are drawn from non-target clean data so that the forget set never removes clean examples of the class the attacker wants to protect, which would confound removal with class deletion. Both A and B are sampled without replacement with a seed derived from (r, p, seed), so the construction is deterministic and reproducible. Algorithm 1 states the procedure.

The construction has three properties that matter for interpretation. First, $|\mathscr{F}|$ grows as p falls at fixed r: with $N_p = 2{,}500$, the cell $(r,p) = (0.90, 1.0)$ removes 2,250 samples, while $(0.90, 0.7)$ removes 3,214. Precision is therefore also a measure of how much clean data the method must give up. Second, r and p are requested values; the realised values are stored with each run and differ only by rounding, because the counts are integers. Third, the construction is deliberately kinder than a real detector in one respect: it never places a false positive in the target class and it never places poison in the retain set when $r = 1$. Real detectors do both. The grid is therefore a best-case controlled manipulation of detector error,

not a simulation of a specific detector. Section VI anchors it to two real detectors by running the same removal methods on their raw flag sets.

D. Defence Success and the Tolerance r^*

A removal run is a success when both halves of the following criterion hold:

$$\mathrm{ASR}(f') \le 5\% \quad \text{and} \quad \mathrm{Acc}_{\mathrm{clean}}(f') \ge \mathrm{Acc}_{\mathrm{clean}}(f_{\mathrm{gold}}) - 2. \tag{5}$$

The first half says the backdoor is no longer usable at a rate that would matter to an attacker; the second says the cure did not cost more than two points of utility. The two-point budget is a modelling choice; it is wide enough to absorb seed noise (the gold model itself varies by about ± 0.1 points) and narrow enough to exclude methods that remove the backdoor by damaging the network. We report the two quantities separately in every table so that a reader who prefers a different budget can re-derive the verdicts. For each method we then define

$$r^* = \min\Big\{ r : \overline{\mathrm{ASR}}(r,1) \le 5\% \text{ and } \overline{\mathrm{Acc}}_{\mathrm{clean}}(r,1) \ge \overline{\mathrm{Acc}}_{\mathrm{clean}}(f_{\mathrm{gold}}) - 2 \Big\}, \tag{6}$$

where the bar is the mean over the three seeds, and write $r^* = -$ when no recall in the grid succeeds. We use the mean because the criterion describes a deployment, not a single training run; the per-seed counts are reported separately in Table VI so that a reader who requires unanimity can see exactly where the two readings diverge. The tolerance is a property of a removal method, not of a detector: it states how much detector recall the method needs before it can be relied on.

E. Secondary Metrics

For the amplification study we ask whether partial removal makes missed poison separable from clean data. Let $\mathscr{M} = \mathscr{P} \setminus \mathscr{F}$ be the poison missed by the forget set and let $\mathscr{C}_0$ be clean training data of the target class. We fit a detector score to distinguish $\mathscr{M}$ from $\mathscr{C}_0$ and report the area under the ROC curve,

$$\mathrm{AUC} = \Pr[s(x_{\mathrm{missed}}) > s(x_{\mathrm{clean}})], \tag{7}$$

with ties counted at one half. An AUC of 0.5 is chance. For durability we report ASR after k gradient steps on fresh poison, a loss-threshold membership inference AUC following Shokri et al. [34], and the Kullback–Leibler divergence between the output distributions on clean and trigger-stamped inputs. A live backdoor shows a large clean-versus-trigger divergence; a dormant one does not.

F. Statistical Treatment

Every CIFAR-10 cell is run with seeds 0, 1, and 2. Poison placement, weight initialisation, and data order vary with the seed; the training and test splits do not. We report mean $\pm$ standard deviation over seeds and, for the success criterion, the count of seeds that pass. With three seeds the counts are coarse, and we do not attach confidence intervals to them. When a method is bimodal across seeds (NegGrad+ is the clearest case), the mean and standard deviation describe a mixture rather than a stable operating point; we say so in the table notes instead of hiding it in a single number.

IV. Removal Methods and Detector Models

All removal methods start from the poisoned model f except the two retraining methods, which start from random initialisation. The post-training methods use SGD with momentum 0.9 and weight decay 5×10^{-4}, and are deliberately kept simple: the goal is to compare families under a common budget, not to tune each family to its best. The CIFAR-10 settings common to all runs are listed in Table I.

A. Deletion and Retraining

RETRAIN$_{\setminus F}$ is the reference deletion method: train a new network from scratch on $\mathscr{D} \setminus \mathscr{F}$ with the full twelve-epoch schedule. It is expensive, but it is the method that the oracle literature reports as a success, and it isolates the effect of the forget set from any property of the poisoned initialisation. When $\mathscr{F}$ equals the true poison set, this is the gold model.

B. Fine-Tuning on the Retained Set

FINE-TUNE continues training the poisoned model for three epochs at learning rate 0.01 on $\mathscr{D} \setminus \mathscr{F}$. It is the cheapest possible response to a detector and is common in practice when retraining is not affordable. It has no mechanism for forgetting: it only reinforces the retained data.

C. Gradient Ascent and SCRUB

NEGGRAD+ performs three epochs of ascent on the cross-entropy of the forget set, balanced by descent on a matched random batch from the retained set, at learning rate 10^{-3} [22]. SCRUB first fine-tunes a teacher on the retained set for three epochs, then trains a student for three epochs to minimise cross-entropy on the retained set plus a Kullback–Leibler term that pulls the student's distribution on the forget set towards the teacher's [25]. Both methods are designed for exact forget sets; the grid tests how they behave when the set is noisy.

D. Repair and Inoculation

The recipe is the only method in the suite that is designed around an imperfect forget set. It has two stages.

Label repair. Features are extracted from the penultimate layer of the poisoned model for all training samples. Each flagged sample receives a pseudo-label by a weighted k-nearest-neighbour vote (k=10, inverse-distance weights) over the *unflagged, non-target* training data. The sample is kept, not deleted, with its repaired label. The intuition is that a poisoned image whose patch is ignored looks like its true class, so the nearest neighbours in feature space are clean images of that class rather than of the target class.

Inoculation. A pool of 5,000 retained samples that the poisoned model does not confidently classify as the target class is trigger-stamped and kept with its true label. Each mini-batch is composed of 60% retained samples, 20% flagged samples with pseudo-labels, and 20% inoculated samples. The network is fine-tuned for three epochs at learning rate 0.01. Presenting the trigger with the correct label teaches the network that the trigger is not predictive of the target class, which is the

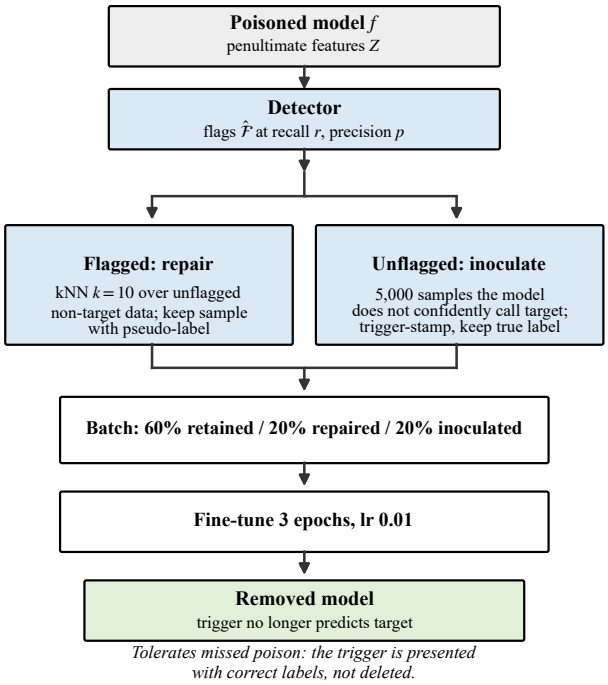


Fig. 2. The repair-and-inoculate recipe. Flagged samples are relabelled by a k-nearest-neighbour vote over unflagged, non-target data; unflagged samples supply the trigger-stamped inoculation pool. The streams are mixed 60/20/20 and fine-tuned for three epochs. The lower branch is what tolerates missed poison: the trigger is presented with correct labels rather than deleted.

mechanism by which the recipe can tolerate poison that the forget set missed. Fig. 2 draws the two streams and where they meet.

Inoculation only. To separate the two stages we also run an inoculation-only variant that keeps the entire training set, applies no label repair, and fine-tunes with 20% trigger-stamped samples for three epochs. This is the F-independent baseline of the E1 grid: it uses no forget set at all and therefore cannot degrade with detector recall, but it also cannot repair the poisoned labels. A second variant replaces the true trigger with a synthesised one (Section VII).

E. Trigger Synthesis

The synthesis baseline follows Neural Cleanse [14]. For each class other than the target, a soft mask $m \in [0,1]^{32\times32}$ and a pattern are optimised for 200 Adam steps at learning rate 0.1 on 64 samples of that class, with the objective of driving the model to the target class while penalising the mask mass with weight 10^{-3}. The class with the smallest mask mass is retained as the estimated trigger. The overlap with the true patch is measured by intersection over union (IoU) after thresholding the mask at 0.5. The synthesised trigger is then used wherever the true trigger would be used, both for inoculation and as the proxy in Calibrate-then-Remove.

F. Calibrate-then-Remove

Calibrate-then-Remove (CaR) is our attempted mechanism for turning detector output into a removal decision with a guarantee. For a grid of spectral-signature thresholds θ, it forms the corresponding forget set, applies the recipe's repair stage to obtain a proxy-removed model, and measures the proxy model's response τ to the *synthesised* trigger. If a threshold yields $\tau \leq 5\%$, the same threshold is applied to the real test trigger and the removal is accepted. If no threshold qualifies, the method falls back to the F1-optimal threshold. The logic is conformal in spirit: the synthesised trigger plays the role of a calibration sample that is exchangeable with the test trigger [32], [33]. It fails if the synthesised trigger is not a valid proxy, which is what we observe.

G. Amplify-and-Detect

The second attempted mechanism tests a hypothesis rather than a pipeline: that partial unlearning makes the poison it failed to remove easier to detect, because the network must reallocate the trigger association onto the residual poison and therefore represents it more sharply. We operationalise it by applying partial label repair at two strengths, then fitting a detector to separate the missed poison from clean target-class data and measuring AUC (7). The pre-registered promotion gate required an AUC gain of at least $+0.10$ over the base model, a clean-accuracy loss of at most two points, and a gain larger than a matched clean-removal control that receives the same intervention on clean data instead of poison. All three clauses had to hold on all seeds.

H. Detectors

We use two published detectors, both applied to the penultimate features of the poisoned model.

Activation clustering [17] processes each class separately: it projects the class's features to ten principal components, runs k-means with k=2, and flags the smaller cluster if its size fraction is below 0.35 and the silhouette score exceeds 0.10.

Spectral signatures [18] computes, for each class, the top singular vector of the centred feature matrix and scores every sample by the squared projection onto that direction. It flags a fixed budget of $1.5N_p$ samples with the largest scores. The fixed budget is why its precision degrades at low poison rates: at a 1% poison rate the budget is 750 samples for 500 true poison, and at 5% it is 3,750 for 2,500.

Fig. 3 draws the two pipelines side by side. Both read the same penultimate features and differ only in how they score a sample; the two annotations mark the failure modes that recur in the grid, an empty flag set at 10% poison for activation clustering and a 33.8% recall at 1% for spectral signatures.

Both detectors were also run on VGG-16-BN and MobileNetV2 models. Those runs were degenerate (no flags, or flags with no true positives) and are excluded from the tables; we report the ResNet-18 operating points, which are the ones that produce non-trivial forget sets.

I. Efficiency-Oriented Baselines

Two further baselines are included for completeness. SISA-5 shards the training set into five parts, trains one model per shard, and unlearns by retraining the shards that contain

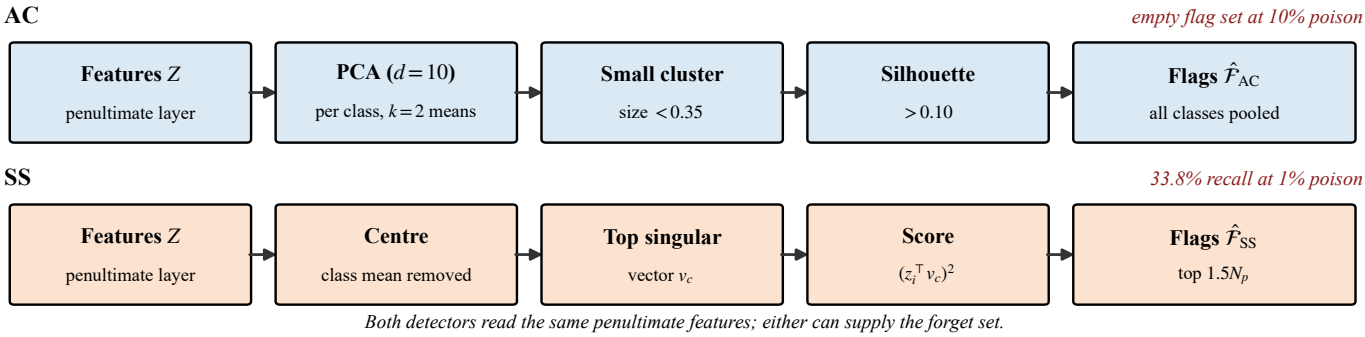


Fig. 3. The two detectors as pipelines. Both read the same penultimate features. Activation clustering (top) returns no flags at a 10% poison rate; spectral signatures (bottom) recovers 33.8% of the poison at a 1% rate because its fixed $1.5N_p$ budget lets clean samples crowd out poison.

forget-set samples [21]; we report both the oracle-recall variant and a recall-0.5 variant. Fine-pruning removes filters with the smallest average activation on clean data, in increasing fractions [19]. These methods are not swept over the grid because their compute profile is different, but they answer the natural question of whether a cheaper mechanism can substitute for deletion.

J. What Each Family Needs

Fig. 4 arranges the families by what they consume. Deletion is the only method that never touches the poisoned weights; it discards the network and pays a full schedule. Fine-tuning and the two ascent methods all start from the poisoned model and differ only in how they use $\mathscr{F}$, and all three assume that $\mathscr{F}$ is the poison, which is the assumption the grid relaxes. The recipe is the only family that consumes the trigger, and that is the asymmetry the figure is meant to make visible: trigger knowledge buys tolerance to an incomplete $\mathscr{F}$, and nothing else in the suite has it. The compute column also explains why the efficiency-oriented methods are evaluated separately rather than inside the grid: deletion buys its clean removal with a full schedule, while the three-epoch methods are cheap precisely because they stay close to the poisoned model.

V. Experimental Protocol

A. CIFAR-10

We use CIFAR-10 [35]: 50,000 training and 10,000 test images of 32×32 pixels in ten classes, with class 0 as the attack target. Poisoning is dirty-label: a fraction of non-target training images receive the trigger and their labels are set to the target. BadNets stamps a 4×4 checkerboard of alternating 1/0 pixels on rows and columns 27–30 [1]. Blend replaces the image by $0.8x + 0.2\pi$, where π is a fixed uniform pattern generated with seed 123 [3]. The poison rate is 5% (2,500 samples) except in the amplification study, which uses 10% (5,000 samples) to give the detector more signal.

We train CIFAR variants of ResNet-18 [36] and VGG-16-BN [37] for 12 epochs with SGD (Nesterov momentum 0.9, weight decay 5×10^{-4}), a OneCycle schedule with maximum learning rate 0.1 (0.05 for VGG), batch size 512, random crop to 28×28 with padding 4, horizontal flip, and mixed precision [38], [39], [40]. Normalisation is applied inside the model. Three seeds (0, 1, 2) vary poison placement, initialisation, and data order; the train/test split is fixed. Each seed therefore yields its own gold model, and success is always judged against that seed's gold accuracy (88.59, 88.65, and 88.77% for seeds 0–2 respectively). The total CIFAR compute was approximately 23 GPU-hours on two T4 cards.

Attack success is measured on the non-target test images after stamping the trigger, so a model that predicts the target class indiscriminately cannot score well by accident. Clean accuracy is measured on the unmodified test set.

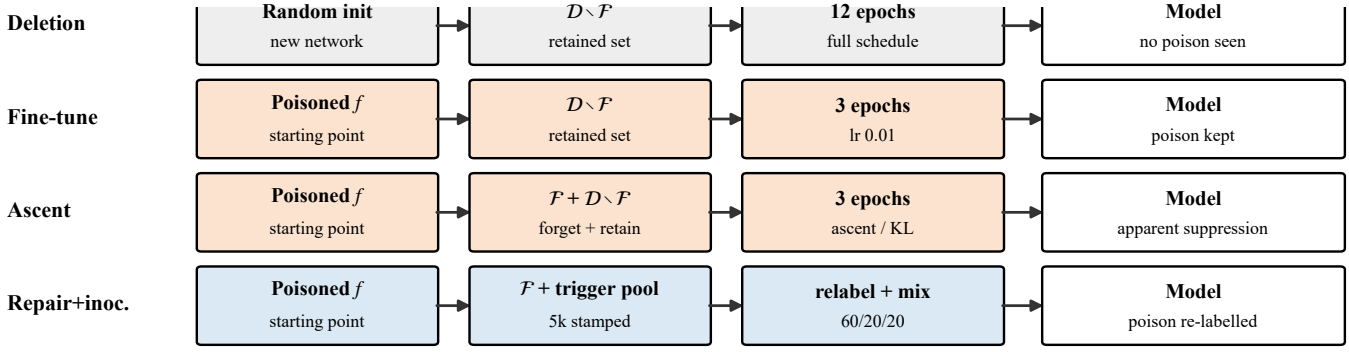
B. RadioML Automatic Modulation Classification

The cross-domain study uses RadioML 2016.10a [41], [42]: eleven modulation classes, SNR from -20 to 18 dB in 2 dB steps, each sample a 2×128 in-phase/quadrature window. Samples are normalised to unit energy. The classifier is a three-block one-dimensional CNN (32, 64, and 128 channels, batch normalisation, max pooling, global average pooling) trained for 12 epochs with Adam at learning rate 10^{-3} and weight decay 10^{-4}. The trigger is a fixed burst that sets $I{=}0.9$ and $Q{=}{-}0.9$ on the last eight samples, and 5% of the non-target training windows are stamped and relabelled to BPSK. The split is 80/20 and fixed across seeds; poison placement and initialisation vary. Removal methods are fine-tuned for four epochs at learning rate 10^{-4}. Four methods (gold retrain, retain fine-tuning, recipe, and inoculation-only) are run at recall 1.0, 0.75, and 0.5 with precision 1.0, plus a precision-0.7 cell at recall 0.75. Attack success and clean accuracy are evaluated per SNR bin.

We emphasise one protocol detail because it changes how the results must be read. At SNR below about -12 dB the clean classifier is close to chance (9.1% for eleven balanced classes), and an ASR threshold of 5% is then not a meaningful test of a defence: even the oracle retrain sits above it. We therefore report γ^* for every SNR but mark the near-chance region and restrict interpretation to SNR at or above -6 dB, where clean accuracy is at least 48%.

C. What Is Frozen

The four experiment drivers were frozen before the final runs and were not edited afterwards. Each driver writes one CSV row per run, including the requested and realised recall



Only the last family uses the trigger; the other three depend on the forget set being complete.

Fig. 4. The four removal families and what each consumes. Only the repair-and-inoculation family uses the trigger; the other three assume that the forget set is the poison. The last column states the intended mechanism, not the measured outcome.

and precision, the forget-set size, clean accuracy, ASR, wall-clock time, and free-text notes. All figures and tables in this paper are generated from those rows by a single analysis script; no number was transcribed by hand. Where a run produced a degenerate detector output (no flags, or flags containing no true poison) the row is kept in the records and excluded from the corresponding table with a note. The logs are keyed by person, experiment, architecture, attack, poison rate, seed, requested recall and precision, method, and detector, so any table can be reconstructed by filtering the logs; the analysis script applies those filters and writes the LaTeX tables and the PDF figures directly. Fig. 5 traces the chain from the frozen core to the compiled PDF; because the analysis is a deterministic function of the CSVs, every table can be regenerated without rerunning a model.

D. Scope of the Evidence

Table II lists the experiment blocks, what each was run for, and where it is reported. The CIFAR grid covers two architectures, two attacks, three seeds, five removal families, and ten recall–precision cells; the amplification study adds a third detector operating point at 10% poison. The radio study covers one architecture, one attack, three seeds, four methods, and twenty SNR bins. Two blocks that were planned in the protocol were not executed: an adaptive-attacker study (E4) and a full hyperparameter sweep for the baselines. We state this explicitly in Section XII rather than implying coverage we do not have. The claims in this paper are correspondingly bounded: they concern post-training removal under the stated attacks and budgets, not the impossibility of defence in general.

TABLE I
FROZEN EXPERIMENTAL PROTOCOL. EVERY ROW OF EVERY TABLE IN THIS PAPER WAS PRODUCED UNDER THESE SETTINGS; ONLY THE LISTED KNOBS VARY BETWEEN EXPERIMENTS.

Item	Setting
Dataset	CIFAR-10, 50k train / 10k test, 10 classes, target class 0
Attacks	BadNets 4×4 checkerboard (rows/cols 27–30); Blend $0.8x + 0.2$ pattern, seed 123
Poison rate	5% (2,500 samples); 10% in the amplification study
Architectures	ResNet-18, VGG-16-BN (CIFAR variants)
Optimiser	SGD, momentum 0.9 (Nesterov), weight decay 5×10^{-4}
Schedule	OneCycleLR, max lr 0.1 (VGG 0.05), 12 epochs, batch 512
Augmentation	random crop 32→28 with padding 4, horizontal flip; normalisation inside the model
Seeds	0, 1, 2 (identical splits; poison placement and init vary)
Detectors	Activation Clustering; Spectral Signatures (per-class, PCA/2-means resp. top singular vector)
Forget set	$\mathscr{F}(r,p)$: $N_p r$ true poison + $N_p r(1-p)/p$ clean non-target false positives
Defence success	ASR $\leq 5\%$ *and* clean accuracy within 2 points of the gold retrain
Hardware	Kaggle 2×T4 (16 GB each), mixed precision; ≈23 GPU-hours total

VI. The Recall Cliff

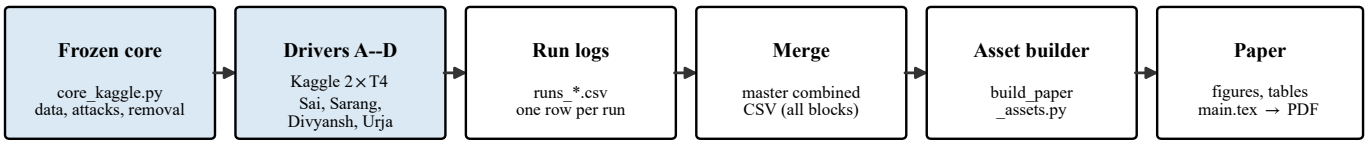
A. Detector Operating Points

Table III and Fig. 6 report the two detectors on ResNet-18/BadNets models at poison rates of 1%, 5%, and 10%. Three features of these measurements motivate the rest of the paper.

First, recall is not free. Spectral signatures flags a fixed budget of $1.5N_p$ samples and recovers only 33.8% of the poison at a 1% poison rate, because the top-scoring clean samples crowd out poison at that scale. Activation clustering recovers all of it at 1% and 5%, but at 10% it returns an empty flag set: the class-0 cluster no longer passes its size and silhouette tests, so the detector reports that nothing is wrong at exactly the poison rate where the attack is strongest.

Second, precision degrades as recall improves. At 5%, activation clustering flags 4,140 samples for 2,500 true poison (precision 0.60) and spectral signatures flags 3,750 (precision 0.64). A removal method therefore receives a forget set in which roughly one third of the entries are clean images.

Third, architecture matters. The same detectors on VGG-16-BN and MobileNetV2 produced degenerate outputs (no flags, or flags with no true positives) and are excluded.



The core is never edited between drivers: seeds, epochs, learning rates and batch size are fixed; every number in the paper is read from a CSV, none is transcribed.

Fig. 5. The artifact chain. Four frozen drivers write one CSV row per run, the rows are merged into a master table, and the asset builder reads that table to write the LaTeX tables and PDF figures in this paper. No number is typed by hand.

TABLE II
EXPERIMENT BLOCKS. E4 WAS PLANNED AND NOT EXECUTED; EVERY OTHER BLOCK IS REPORTED IN THE SECTION LISTED.

Block	Question	What was run	Sec.
E1	Does removal survive imperfect recall?	10 (r,p) cells × 5 methods × 3 seeds, ResNet-18/BadNets; inoculation-only baseline	VI
E2	Is the cliff about residual poison?	4 architecture–attack pairs × 5 residual fractions (0–100%)	VI
E3	Can calibration choose the threshold?	20 SS thresholds with proxy removal; two full retrains	VIII
E4	Does an adaptive attacker break the recipe?	planned, not executed	XII
E5	Erasure or suppression?	4 settings × relearning to 50 steps; MIA and KL probes	IX
E6	Do real detector columns reproduce the cliff?	AC and SS at 1%, 5%, 10% poison; deletion and recipe at realised operating points	VI
E7	Does the cliff transfer?	VGG-16-BN/BadNets, VGG-16-BN/Blend, ResNet-18/Blend at r=0.75, 0.5	VII
E7b	Can the trigger be synthesised?	Neural-Cleanse synthesis on two attacks	VII
E8	Is ascent removal real?	4 ascent strengths × raw and BN-guarded	VIII
E9	Does partial repair amplify detectability?	10% poison, 3 seeds × 2 repair strengths + matched controls	VIII
SISA	How do efficiency-oriented baselines compare?	SISA-5 shards; fine-pruning at two thresholds	IX
RF	Does the cliff cross domains?	RadioML AMC, 4 methods × 3 recalls, 20 SNR bins, 3 seeds	X

TABLE III
MEASURED DETECTOR OPERATING POINTS: THE NOISY FORGET SETS THAT MOTIVATE THE GRID.

Poison	Det.	Flagged	TP	Precision	Recall
1%	AC	507	500	0.986	1.000
1%	SS	750	169	0.225	0.338
5%	AC	4,140	2,500	0.604	1.000
5%	SS	3,750	2,411	0.643	0.965
10%	AC	0	0	–	–
10%	SS	7,500	2,400	0.320	0.480

Detector output on a ResNet-18/BadNets model; TP is the number of true poison samples among the flags. At 10% poison, AC returns an empty flag set (no cluster passes its size/silhouette test), i.e. $r = 0$: the detector, not the removal method, is the binding constraint. VGG-16-BN and MobileNetV2 detector runs were degenerate (no flags or no true positives) and are excluded.

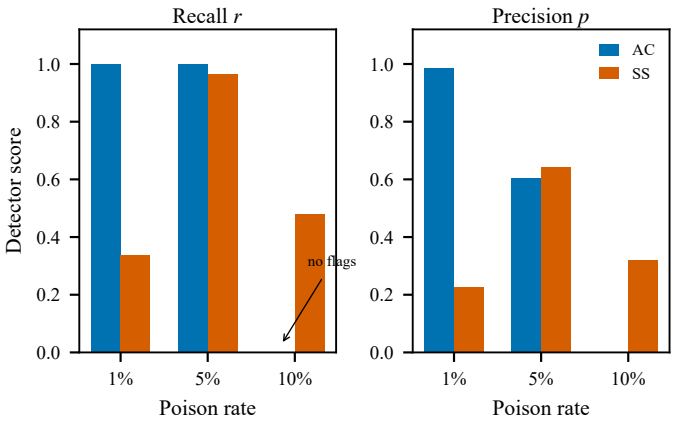


Fig. 6. Detector recall and precision on ResNet-18/BadNets at three poison rates. Activation clustering (AC) returns no flags at 10% poison; spectral signatures (SS) is limited by its fixed $1.5N_p$ flag budget at 1%.

Detector behaviour is not a property of the attack alone; it is a property of the representation, which is why a controlled parameterisation of detector output is more portable than a fixed threshold.

B. Deletion Collapses Between r=0.95 and r=0.90

Fig. 7 and Table IV give the central result. Retraining on the retained set is essentially perfect when the forget set contains every poisoned sample: ASR $1.68 \pm 0.26\%$ at r=1, with clean accuracy 88.67% against a gold value of 88.59–88.77%. It remains low at r=0.95 (ASR $2.84 \pm 0.73\%$), but the clean accuracy of that model is 84.58%, four points below gold, so the joint criterion already fails. At r=0.90 the backdoor is fully back: ASR $100.00 \pm 0.00\%$ with clean accuracy 88.80%. From there down to r=0.25 the ASR stays between 99.63 and 100.00%. The transition is not gradual. Removing 2,375 of 2,500 poison samples leaves a model that still cannot be attacked at a useful rate; removing 2,250 does not.

The r=0.95 cell deserves a comment because it looks like a counterexample. No clean samples are removed there (p=1),

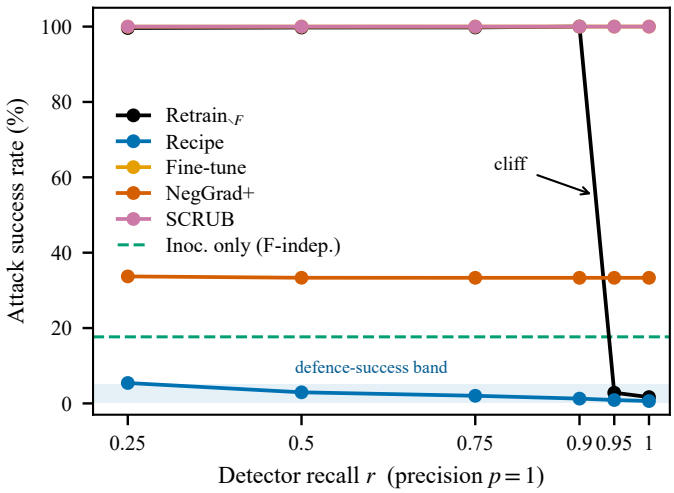


Fig. 7. Attack success after removal as a function of detector recall at precision 1.0 (mean over three seeds). Deletion is flat at 100% for $r \leq 0.90$ and drops only when every poisoned sample is removed. The dashed line is the inoculation-only baseline, which uses no forget set; the shaded band is the 5% success threshold.

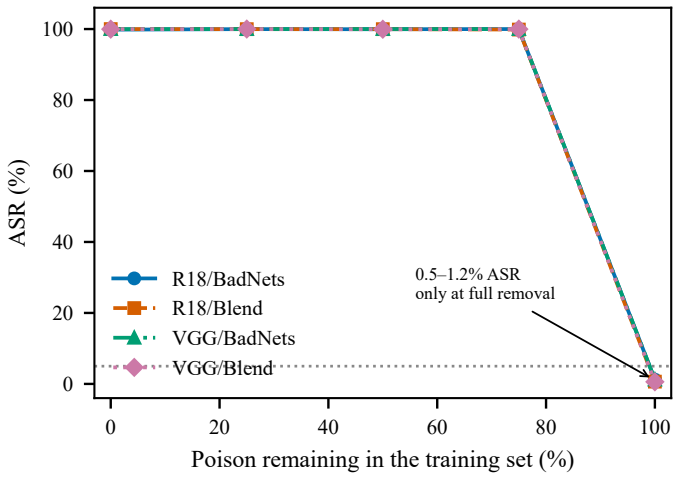


Fig. 8. ASR versus the fraction of poison remaining in the training set, for four architecture–attack combinations. The curve is flat until the poison is completely gone.

so the four-point clean-accuracy drop cannot be a false-positive effect. It is a property of retraining on a set that still contains 125 poison samples, and we did not isolate the mechanism. We report it as measured; the practical consequence is the same, because the joint criterion is what a deployment decision would use. It is also the reason r^*=1.00 rather than 0.95: the ASR cliff is at 0.90, but the success cliff is at 1.00.

C. The Cliff Is a Function of Residual Poison

A separate block (E2) varies the fraction of poison left in the training set directly, without a detector and without simulated false positives. Table V and Fig. 8 show the result for the four architecture–attack combinations. With 0%, 25%, 50%, or 75% of the poison still present, ASR is between 99.83% and 100.00% in every cell. At complete removal it falls to 0.59–1.22%. Clean accuracy varies by at most 0.9 points across the five fractions within each setting, so the collapse is not a side-effect of a damaged network. The shape is a step, and the step is located at zero residual poison. This is the same phenomenon as the grid's r=1 boundary, seen from the other side: what matters is not how much poison was deleted but whether any remains. The absolute clean accuracies in E2 are 89.2–90.5%, slightly above the E1 gold values, because the block was trained separately; the ASR pattern, not the absolute accuracy, is the object of comparison.

D. Precision Is Secondary to Recall

The precision axis is much flatter than the recall axis. At fixed r=0.90, moving from p=1.0 to p=0.9 or p=0.7 enlarges the forget set from 2,250 to 2,500 or 3,214 samples, and deletion's ASR moves from 100.00% to 99.99% in both cases. At fixed r=0.50, p=0.7 gives 99.90% against 99.76% at p=1.0. For the recipe the same manipulation is visible but small: ASR goes from 1.27% to 1.90% at r=0.9 and from 2.95% to 2.59% at r=0.5, with no consistent direction. Deleting extra clean data costs some accuracy (the recipe's clean accuracy moves within about 0.6 points), but it neither rescues nor destroys a removal that is otherwise working.

The asymmetry has a simple reading. A false positive is a clean sample removed from training, which the network can absorb if the dataset is large. A false negative is a poisoned sample kept in training, which re-installs the association the removal was meant to break. At a 5% poison rate, the largest precision penalty we test (p=0.7) adds 714 clean samples to a 50,000-sample training set, a 1.4% perturbation. The smallest recall penalty we test (r=0.95) leaves 125 poison samples, which is enough to matter. Detector engineering for removal should therefore optimise recall first and treat precision as a budget constraint.

E. The Other Method Families

The remaining families do not merely fail at low recall; most of them fail at full recall as well.

FINE-TUNE never removes the backdoor: ASR is $100.00 \pm 0.00\%$ at r=1 and above 99.96% in every other cell, while clean accuracy is unaffected (89.02%). Three epochs on the retained set reinforce whatever the poisoned model already represents.

NEGGRAD+ is bimodal. In most cells the mean ASR is 33.33% with a standard deviation of 57.74, which is the signature of two seeds near zero and one seed at 100%. Where it does suppress the backdoor, it does so by damaging the model: clean accuracy is $79.90 \pm 7.55\%$ at r=1, and at $(r, p) = (0.9, 0.9)$ it is $54.23 \pm 38.31\%$. Under the joint criterion it never succeeds, and the bimodality means that a single-seed evaluation of this method would be uninformative in either direction.

SCRUB leaves ASR at $100.00 \pm 0.00\%$ in every one of the ten cells. Its clean accuracy is also unstable, from $65.76 \pm 20.17\%$ at r=1 to $86.07 \pm 2.59\%$ at r=0.25. The teacher-student objective assumes that the forget set is the poison; when it is not, the student is trained to match the teacher on poison and the backdoor survives.

TABLE IV
E1: THE RECALL–PRECISION GRID. DELETION MEETS THE SUCCESS CRITERION ONLY AT FULL RECALL; REPAIR+INOCULATION HOLDS ASR BELOW 5% DOWN TO r=0.5.

r	p	$\lvert\mathscr{F}\rvert$	**Retrain$_{\setminus F}$**	**Fine-tune**	**NegGrad+**	**SCRUB**	**Recipe**
1.00	1.0	2,500	**1.68±0.26**	100.00±0.00	33.33±57.74	100.00±0.00	**0.63±0.21**
0.95	1.0	2,375	**2.84±0.73**	100.00±0.00	33.33±57.74	100.00±0.00	**0.90±0.11**
0.90	1.0	2,250	100.00±0.00	100.00±0.00	33.33±57.74	100.00±0.00	**1.27±0.58**
0.75	1.0	1,875	99.78±0.22	100.00±0.01	33.33±57.74	100.00±0.00	**2.03±1.32**
0.50	1.0	1,250	99.76±0.31	100.00±0.01	33.35±57.72	100.00±0.00	**2.95±0.78**
0.25	1.0	625	99.63±0.58	99.97±0.02	33.71±57.41	100.00±0.00	5.42±1.82
0.90	0.9	2,500	99.99±0.01	100.00±0.00	33.33±57.74	100.00±0.00	**1.16±0.43**
0.90	0.7	3,214	99.99±0.02	100.00±0.00	33.33±57.74	100.00±0.00	**1.90±0.34**
0.50	0.9	1,389	99.77±0.21	99.97±0.03	33.35±57.72	100.00±0.00	**3.57±0.40**
0.50	0.7	1,786	99.90±0.15	99.99±0.01	33.34±57.73	100.00±0.00	**2.59±1.03**

ASR (%) after removal, mean±s.d. over three seeds; **bold** marks mean ASR $\leq 5\%$ (the ASR half of defence success; clean accuracy is checked in Table VI). $\lvert\mathscr{F}\rvert$ is reported for the seed-0 draw (identical across seeds up to rounding). NegGrad+ is bimodal across seeds: two seeds drive ASR to ≈ 0 while collapsing clean accuracy to 75–86% at p=1 (in the four $p < 1$ cells one of those seeds collapses further to 10%), one stays fully inert (ASR 100%); the mean and s.d. describe the mixture, not a stable operating point.

TABLE V
E2: THE CLIFF IS A STEP FUNCTION OF RESIDUAL POISON, NOT OF THE AMOUNT DELETED.

Setting	**0%**	**25%**	**50%**	**75%**	**100%**	**Gold**
R18/BadNets	99.83	99.98	99.94	100.00	1.21	90.3
R18/Blend	99.99	99.97	99.97	99.86	0.64	90.2
VGG/BadNets	100.00	99.99	100.00	100.00	1.22	89.7
VGG/Blend	99.97	100.00	99.94	99.98	0.59	89.7

ASR (%) versus the fraction of poison left in the training set after deletion. The last column is the gold retrain's clean accuracy for that setting; across the five residual fractions, clean accuracy varies by at most 0.9 points in every setting, so the cliff is not an accuracy artefact.

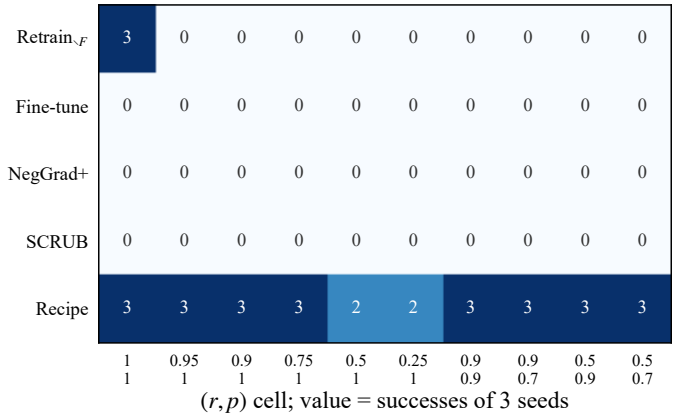


Fig. 9. Number of seeds (of three) meeting both success criteria in each (r,p) cell. Deletion is dark everywhere except the full-recall corner; the recipe is dark everywhere except the two lowest-recall cells at p=1.

Inoculation only is the interesting baseline. It uses no forget set at all, so it cannot degrade with recall, and it does reduce ASR from near 100% to 17.67% with clean accuracy 88.41%. That is a real effect, but it is above the 5% threshold and therefore not a success by our criterion. It also shows that inoculation alone is not sufficient on CIFAR-10; the combination with label repair is what produces the recipe's numbers.

The recipe holds ASR between 0.63 and 2.95% for every cell with $r \geq 0.5$ and at p=1, and between 1.16 and 3.57% for the $p < 1$ cells, with clean accuracy between 87.37 and 88.00%. It degrades at r=0.25 (5.42 ± 1.82%) and is the only method whose failure, when it comes, is a gradual loss of margin rather than a return of the backdoor. The reason is structural: its inoculation stage does not require the forget set to be complete. We return to the cost of that property in Section VII.

F. Success Counts and the Tolerance r^*

Table VI and Fig. 9 convert the grid into the joint criterion. Deletion succeeds on all three seeds in exactly one cell, $(r,p) = (1,1)$, and on no seed anywhere else; its tolerance is r^*=1.00. Fine-tuning, NegGrad+, and SCRUB have no successful cell. The recipe succeeds on 3/3 seeds in eight of the ten cells and on 2/3 in the two r=0.5 and r=0.25 cells at p=1. At r=0.5 one seed misses the clean budget by 0.09 points. At r=0.25 one seed exceeds the ASR threshold (7.49%), and the three-seed mean does too (5.42%), so that cell is outside the mean rule as well. At $(0.5, 1.0)$ the mean clean accuracy is 87.37% against a gold mean of 88.67%, which is inside the two-point budget, so the mean definition of Eq. (6) gives r^*=0.50. An all-seeds rule would instead give 0.75, and Table VI reports the counts so that both readings are visible.

G. The Grid in Four Views

Fig. 10 rearranges the ten cells along four axes. Panel (a) follows clean accuracy across recall at precision 1.0. Deletion sits on the gold line at every recall except r=0.95, where it loses 4.09 points, and r=1, where it *is* the gold; SCRUB's damage is concentrated at full recall, where its three seeds are 57.05%, 88.83%, and 51.41% (mean 65.76%); the recipe stays within 1.3 points across the whole range. Panels (b) and (c) are the precision axes at r=0.9 and r=0.5; both are almost flat. Moving p from 1.0 to 0.7 changes deletion's ASR by at most 0.3 points and the recipe's by at most 0.7, while fine-tuning and SCRUB remain at 100% and NegGrad+ near its bimodal mean of 33.3%. Panel (d) isolates the one exception: at r=0.5, NegGrad+'s clean accuracy falls from 83.82% at p=1 to 57.78% at p=0.7. A larger forget set gives its ascent step more clean samples to damage. That is the only visible precision effect in the grid, and it belongs to a method that fails at every precision.

TABLE VI
DEFENCE-SUCCESS COUNTS AND THE RECALL TOLERANCE r^*.

Method	Successes (of 3) per (r,p) cell, ordered as in Table IV										r^*
Retrain$_{\setminus F}$	3	0	0	0	0	0	0	0	0	0	1.00
Fine-tune	0	0	0	0	0	0	0	0	0	0	–
NegGrad+	0	0	0	0	0	0	0	0	0	0	–
SCRUB	0	0	0	0	0	0	0	0	0	0	–
Recipe	3	3	3	3	2	2	3	3	3	3	0.50

Number of seeds (of 3) meeting both criteria (ASR $\leq 5\%$ and clean accuracy within 2 points of the gold retrain), in the order of Table IV. r^* is the smallest p=1 recall at which the three-seed mean meets both criteria (Eq. (6)); the counts allow the stricter all-seeds rule to be read off, and under it the recipe's tolerance would be 0.75. At $(0.5, 1.0)$ the recipe's mean clean accuracy is 87.37% against a gold mean of 88.67%, inside the budget, although one seed misses it by 0.09 points. Gold clean accuracy: 88.59 / 88.65 / 88.77.

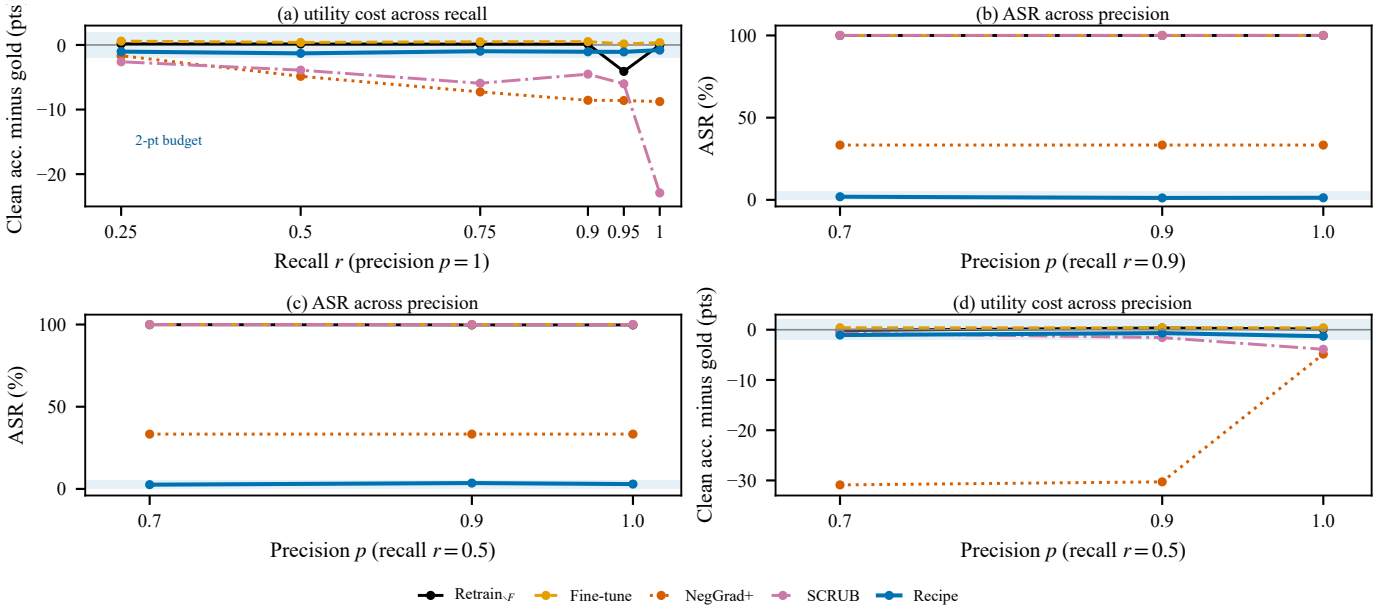


Fig. 10. The E1 grid rearranged. (a) Clean accuracy relative to gold across recall at p=1; the shaded band is the two-point budget. (b), (c) ASR across precision at r=0.9 and r=0.5. (d) Clean accuracy relative to gold across precision at r=0.5. The recall axis moves every method; the precision axes move only NegGrad+.

H. Real Detector Columns

The grid uses simulated forget sets. Table VII checks the conclusion against the detectors' raw flag sets, with no simulation. Activation clustering flags exactly the poison at 5% in this run, and the resulting deletion is a success (ASR 1.62%, clean 88.70%). Spectral signatures recovers 83.4% of the poison at precision 0.834, and the same deletion method fails completely: ASR 99.87% with clean accuracy 88.89%. The SS column uses the detector's raw flag set (2,500 flags) rather than the fixed $1.5N_p$ budget of Table III, so its operating point differs from that table. The recipe succeeds in both columns (0.70% and 1.43%), and SCRUB fails in both (100.00%). This is the cleanest statement of the paper's practical claim: a 16.6% recall shortfall, which would be an unremarkable detector result, is enough to make the strongest deletion method useless.

TABLE VII
E6: REALISTIC DETECTOR COLUMNS REPRODUCE THE SIMULATED GRID.

Method	Recall	Precision	Clean	ASR
Retrain$_{\setminus F}$	1.000	1.000	88.70	1.62
SCRUB	1.000	1.000	61.37	100.00
Recipe	1.000	1.000	88.04	0.70
Retrain$_{\setminus F}$	0.834	0.834	88.89	99.87
SCRUB	0.834	0.834	86.89	100.00
Recipe	0.834	0.834	88.18	1.43

Removal driven by the detectors' *raw* flag sets (no simulation), seed 0. AC flags exactly the poison here; SS trades 16.6% of recall for 16.6% precision. Deletion inherits the detector's recall error directly; repair+inoculation does not.

VII. BEYOND ONE SETTING: ARCHITECTURE, ATTACK, AND TRIGGER KNOWLEDGE

A. The Cliff Transfers

Table VIII repeats the critical cells on three settings: VGG-16-BN with BadNets, VGG-16-BN with Blend, and ResNet-18 with Blend. ResNet-18/BadNets is already covered by the grid. In every setting, deletion at recall 0.75 or 0.5 fails completely: ASR is 99.42–100.00% and clean accuracy is indistinguishable from gold. In every setting, the recipe at the same recall holds ASR at or below 1.02%. The largest clean-accuracy cost is 4.09 points, in the VGG-16-BN/Blend cell at r=0.5 (84.80% against a gold of 88.89%); VGG-16-BN/BadNets costs 3.26 points at the same recall. ResNet-18 is cheaper: 2.60 points for Blend at r=0.75 and 2.04 points at r=0.5. The recipe is thus not free, and its cost grows with the depth of the architecture

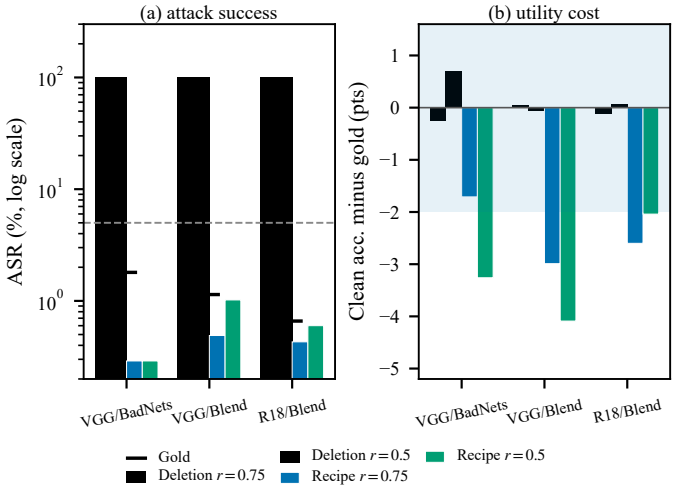


Fig. 11. The transfer settings. (a) ASR after deletion and after the recipe at recall 0.75 and 0.5, on a log scale; the dashed line is 5%. (b) Clean accuracy relative to each setting's gold model; the shaded band is the two-point budget. Deletion stays within 0.7 points of gold and does not remove the backdoor; the recipe removes it at a cost of 1.7–4.1 points.

TABLE VIII
E7: THE CLIFF AND THE RECIPE'S TOLERANCE TRANSFER ACROSS ARCHITECTURE AND ATTACK.

Method	VGG-16-BN BadNets	VGG-16-BN Blend	ResNet-18 Blend
Gold	88.95 / 1.80	88.89 / 1.14	88.75 / 0.66
Deletion, r=0.75	88.71 / 100.00	88.93 / 99.42	88.64 / 99.70
Deletion, r=0.5	89.64 / 100.00	88.83 / 99.84	88.81 / 99.94
Recipe, r=0.75	87.24 / 0.29	85.90 / 0.49	86.15 / 0.43
Recipe, r=0.5	85.69 / 0.29	84.80 / 1.02	86.71 / 0.60

Cells are clean accuracy / ASR (%). Gold is the oracle retrain at r=1. Deletion fails at both recall levels for every architecture–attack pair; the recipe holds ASR $\leq$ 1.02%, at a clean-accuracy cost of up to 4.09 points at r=0.5.

and the strength of the trigger, but it remains inside a plausible budget where deletion is not.

Fig. 11 shows the same cells on a log scale. Panel (a) separates the two outcomes: every deletion cell is at 99.42–100.00%, more than an order of magnitude above the 5% line, while every recipe cell is at 0.29–1.02%. Panel (b) is the price. Deletion is within 0.7 points of gold in every cell, because the poison it fails to remove also does not change clean accuracy at this scale. The recipe's cost is ordered by trigger strength first and depth second: 1.71 and 3.26 points for VGG-16-BN/BadNets at r=0.75 and 0.5; 2.99 and 4.09 for VGG-16-BN/Blend; 2.60 and 2.04 for ResNet-18/Blend. The cost exceeds three points at r=0.5 on VGG-16-BN for both attacks (3.26 points with BadNets and 4.09 with Blend).

The transfer matters because the cliff could have been an artefact of one architecture's optimisation. It is not. The step appears in a network with batch normalisation at every layer and in one with a different depth and width, under a local patch trigger and under a global blend trigger. The location of the step does not move: it lies between full removal and partial removal at 75%, consistent with the E1 boundary between 95% and 90% recall.

TABLE IX
E7B: REPLACING THE TRUE TRIGGER WITH A SYNTHESISED ONE DESTROYS THE RECIPE'S ADVANTAGE.

Setting	Clean	ASR	IoU	Mask mass
R18/BadNets	87.67	98.99	0.000	–
R18/Blend	87.32	95.72	–	166

Inoculation with a synthesised trigger, recall 1.0. The true BadNets patch occupies 16 pixels; the synthesised Blend mask has a soft mass of 166 pixels, about ten times larger. With the true trigger the recipe holds ASR at 0.63% on ResNet-18/BadNets at r=1 (Table IV) and at 0.43–0.60% on ResNet-18/Blend at r=0.75 and 0.5 (Table VIII). IoU and mask mass were not both logged for every cell; the table reports what each run recorded.

B. The Recipe's Tolerance Is Bought With Trigger Knowledge

The recipe's advantage comes from its inoculation stage, which presents the trigger to the network with the correct label. That step requires the trigger. In the grid, and in Table VIII, the true trigger is used. This is a known-trigger upper bound, and it is essential to say so: a defence that must be told the trigger is not a blind defence, and a practitioner who does not know the trigger cannot run the recipe in the form that produces these numbers.

To measure how much the bound is worth, we replace the true trigger with one synthesised from the poisoned model by the Neural Cleanse procedure described in Section IV. Table IX reports the result. On ResNet-18/BadNets the synthesised mask has zero intersection with the true 4×4 patch (IoU $= 0.000$), and inoculation with it leaves ASR at 98.99% with clean accuracy 87.67%. On ResNet-18/Blend the synthesised mask has a soft mass of about 166 pixels, roughly ten times the 16 pixels of the true patch, and inoculation with it leaves ASR at 95.72% with clean accuracy 87.32%. In other words, the synthesis stage recovers a mask that is large and diffuse, not the compact patch that the attack uses, and the inoculation built on it does not remove the backdoor.

This is the strongest negative result in the paper, because it closes the obvious escape route. One might hope that a defence could use the recipe's tolerance without knowing the trigger, by synthesising the trigger first. On these two attacks, that composition fails at the first step. The recipe is therefore best read as a ceiling: it shows what trigger-aware repair can achieve under imperfect recall, and it bounds from above what any trigger-synthesis-based defence could achieve if synthesis were perfect. It does not demonstrate a deployable method.

C. Which Component Does the Work?

The E1 grid contains an inoculation-only baseline (17.67% ASR) and the E8 block contains a label-repair-only baseline (8.63–16.36% ASR, discussed in Section VIII). Neither alone crosses the 5% threshold; the combination does, at 0.63–2.95% across the grid. The two stages are complementary rather than redundant. Repair removes the label noise that would otherwise teach the network to associate the trigger with the target; inoculation changes the association even for poison that never entered the forget set. The first is bounded by recall, the second is bounded by trigger knowledge. The recipe works in the middle ground where each covers the other's weakness.

This decomposition also explains why the recipe degrades gracefully rather than catastrophically. At r=0.5, half the poison is still in the training set, but its labels have been repaired (for the half that was flagged) and the trigger has been re-labelled for the whole network through inoculation. The residual association is weak, and ASR sits at 2–4% rather than at 100%. At r=0.25 the repair covers too little of the poison for the inoculation to dominate, and ASR rises to 5.42%. The transition is gradual because it reflects a balance between two mechanisms, not a threshold in the data.

D. Poison Rate

The amplification study (Section VIII) runs at a 10% poison rate and therefore provides a second scale. At that rate, a partial label repair without inoculation lowers ASR from 99.99% to between 11.07% and 33.03% depending on seed and strength. That is a substantial reduction, but it is not a removal by our criterion, and it occurs in a regime where the detector itself is weakest: activation clustering returns no flags at 10%, and spectral signatures recovers 48.0% of the poison. The combination of a harder attack and a worse detector is exactly the regime in which a controlled study is most useful, because it is where the difference between a method that needs recall and a method that does not becomes visible. We do not claim that the recipe's tolerance is rate-independent; the 10% block does not include the full recipe, and we did not run the full grid at 10%.

VIII. Calibration and Amplification Fail

The grid says that trigger-aware repair is the only family that tolerates imperfect recall, and that it needs the trigger. Two natural responses are to avoid the trigger requirement through calibration and to rescue deletion by making the poison it missed easier to find. We tried both. Neither works under our pre-registered criteria.

A. Calibrate-then-Remove

CaR uses a synthesised trigger as a proxy for the real one. For each spectral-signature threshold θ, it builds the corresponding forget set, runs the recipe's repair stage, and measures the proxy-removed model's response τ to the synthesised pattern. A threshold is acceptable only if $\tau \leq 5\%$, because a model that no longer responds to the proxy is presumed not to respond to the real trigger either. If no threshold qualifies, the method falls back to the F1-optimal one.

Table X reports the sweep. Across twenty thresholds, τ is 100% in nineteen cases and 97.5% in the remaining one. The proxy pattern is optimised on the poisoned model and has zero intersection with the true patch, so label repair does not remove the model's response to it; the calibration signal is saturated regardless of how much poison the threshold captures. The recall of the sweep ranges from 0.003 to 1.000 and its precision from 0.333 to 1.000, so the sweep does cover the useful range. The best F1 is 0.918 at recall 0.894 and precision 0.942, and the model produced there still has ASR 99.97% and clean accuracy 86.71%. The CaR rule never fires, the fallback is taken, and the fallback does not remove the backdoor.

TABLE X
E3: Calibrate-then-Remove fails because the proxy signal is uninformative.

θ	Recall	Precision	τ	F1	Clean	ASR
0.01	1.000	0.333	97.5	0.500	82.32	99.78
20.71	1.000	0.396	100.0	0.567	88.01	100.00
35.33	0.998	0.486	100.0	0.654	88.77	100.00
50.23	0.992	0.628	100.0	0.769	88.88	100.00
74.23	0.956	0.863	100.0	0.907	89.19	100.00
127.71	0.633	0.999	100.0	0.775	89.20	100.00
191.72	0.161	1.000	100.0	0.277	89.29	99.98
291.83	0.003	1.000	100.0	0.006	89.22	99.98
F1-optimal	0.894	0.942	100.0	0.918	86.71	99.97

Calibrate-then-Remove, seed 0. θ is the Spectral-Signature threshold, τ the trigger response of the proxy-removed model (%). Nineteen of twenty thresholds give τ=100% and the twentieth 97.5%: the synthesised trigger never elicits the backdoor (IoU $= 0.000$ against the true patch), so no threshold satisfies $\tau \leq 5\%$ and CaR degenerates to its fallback. Both the F1-optimal and the CaR choice leave ASR $\approx$ 100%.

The failure is instructive. A calibration guarantee is only as good as the exchangeability between the calibration signal and the test signal. Here the proxy is a different pattern, optimised for a different purpose, and the repair stage is not designed to suppress it. The guarantee is vacuous before any threshold is chosen.

B. The Ascent Artefact

A second attempted rescue is to forget harder. Feature-space ascent on the forget set is a standard unlearning baseline, and in our first runs it appeared to work: at learning rate 10^{-3} for two epochs, ASR fell to 0.18% and at four epochs to 0.00% (Table XI, Fig. 12). Clean accuracy fell with it, to 85.29% and then to 77.25%, which we initially read as the usual utility cost of unlearning.

The reading was wrong. Batch normalisation maintains running estimates of activation statistics, and ascent updates them along with the weights. Re-running the identical schedules with the batch-norm statistics frozen to those estimated on the retained set removes the apparent effect entirely: ASR is 99.94–99.98% at every learning rate, with clean accuracy 88.72–88.74% and no utility loss at all. The backdoor was never removed; the normalisation layers were corrupted, and the corrupted statistics made the trigger inoperative while also damaging clean accuracy. A reader who evaluates unlearning without a normalisation control can therefore report a successful removal that is an artefact of the evaluation pipeline.

Label repair at comparable strength behaves differently. At learning rate 0.005 for two epochs ASR is 10.11% with clean accuracy 87.91%; at 0.01 for one epoch it is 16.36% with 87.15%; a re-run of the same cells gave 8.63% and 15.11%. These are genuine reductions, achieved without damaging the network, but they are above the 5% threshold. Repair needs the inoculation stage to cross it.

C. The Amplification Hypothesis

If partial unlearning cannot remove the missed poison, can it at least make that poison easier to detect afterwards? The hypothesis is plausible: if the network must support the

TABLE XI
E8: ASCENT-BASED 'REMOVAL' IS A NORMALISATION ARTEFACT.

Variant	Setting	Clean	ASR
Raw ascent	lr0.0002 ep2	88.03	18.06
Raw ascent	lr0.0005 ep2	87.31	2.87
Raw ascent	lr0.001 ep2	85.29	0.18
Raw ascent	lr0.001 ep4	77.25	0.00
BN-guarded ascent	lr0.0002 ep2	88.74	99.98
BN-guarded ascent	lr0.0005 ep2	88.72	99.98
BN-guarded ascent	lr0.001 ep2	88.72	99.97
BN-guarded ascent	lr0.001 ep4	88.73	99.94
Label repair	lr0.005 ep2	87.91	10.11
Label repair	lr0.01 ep1	87.15	16.36
Label repair (v2)	lr0.005 ep2	87.94	8.63
Label repair (v2)	lr0.01 ep1	87.45	15.11

Seed 1. Raw feature-space ascent appears to remove the backdoor only by destroying the network: clean accuracy falls from 88.03% at the gentlest setting to 77.25% at lr 10^{-3} for four epochs, 11.4 points below the gold mean. Re-running the identical schedule with batch-norm statistics frozen to the retain set leaves ASR at ≈ 100%: the apparent removal was BN corruption, not unlearning. Label repair at the same strength reduces ASR while keeping clean accuracy within 1.5 points.

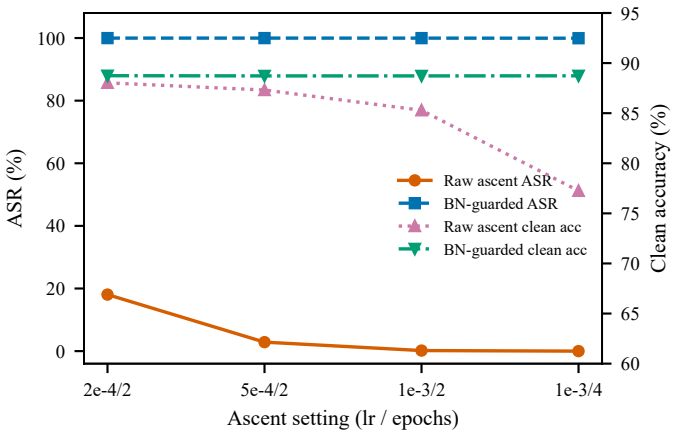


Fig. 12. Raw feature-space ascent appears to remove the backdoor while clean accuracy collapses; the identical schedules with batch-norm statistics frozen to the retain set leave ASR at 100%. The apparent removal is a normalisation artefact.

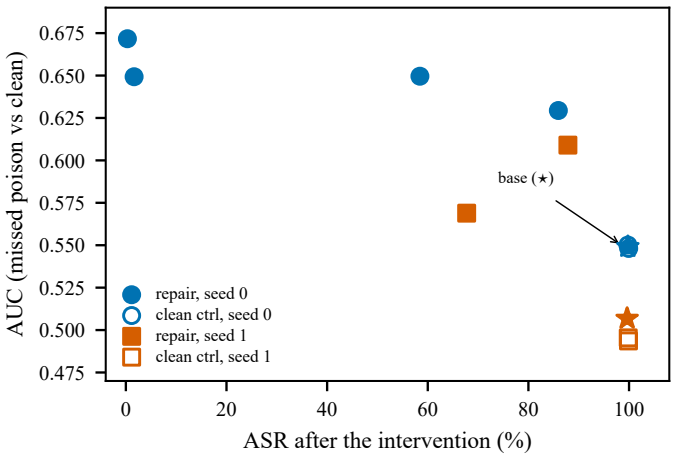


Fig. 13. Detectability of missed poison after partial label repair. Filled markers are repaired models, open markers are matched clean-removal controls, stars are the poisoned base. AUC rises only where clean accuracy has fallen by 2–10 points; the controls stay at chance.

backdoor on a smaller set of poisoned samples, it may allocate more capacity to them, and their representations may separate more cleanly from clean data. If true, the right response to imperfect recall would be an amplify-and-detect loop: partially remove, re-detect the residue, remove again.

We tested the hypothesis with a pre-registered promotion gate. Partial label repair was applied at two strengths on the 5%-poison ResNet-18/BadNets and ResNet-18/Blend models, and a spectral-signature score was fitted to separate the missed poison from clean target-class data. The gate required an AUC gain of at least +0.10 over the base model, a clean-accuracy loss of at most two points, and a gain larger than a matched control that received the same intervention on clean data instead of poison, on all seeds.

Fig. 13 shows the outcome on two seeds. The base models sit at AUC 0.507–0.549 with ASR 99.6–99.8% and clean accuracy 63.5–63.9%. Six repaired models were run. All of them reduce ASR (to between 0.31% and 87.85%) and all raise AUC relative to their base, to between 0.569 and 0.672. But every one of them also loses clean accuracy: the smallest loss is 2.2 points, and the range is 53.9–61.3%. The highest AUC, 0.672, occurs at ASR 0.31% and clean accuracy 55.68%; the next, 0.650, at ASR 58.44% and clean accuracy 59.58%. The matched clean-removal controls, which apply the same procedure to clean data instead of poison, stay at AUC 0.494–0.550 with clean accuracy 62.3–63.3%. The pattern is that of a damaged network whose features separate everything a little better, not of a backdoor made salient: the amplification signal and the utility loss arrive together. No repaired row remains within two points of its base clean accuracy, so the pre-registered gate fails on the utility clause even in the rows where the AUC clause passes.

D. Full-Scale Test

The spike used 5% poison and a partial repair. To test the hypothesis at the scale where it would matter, we repeated it with 10% poison, three seeds, two repair strengths, and matched controls (Table XII, Fig. 14). The base models have AUC 0.494–0.499. Repair at learning rate 0.005 gives 0.470–0.497; repair at 0.01 gives 0.468–0.500. The controls give 0.492–0.503. Every value is at chance, and the largest gain over the base mean of 0.496 is 0.006. It comes from a control, not from a repaired model; the largest repair gain is 0.003. Both are two orders of magnitude below the gate of +0.10.

The repair does reduce ASR, from 99.99–100.00% to 11.07–33.03%, and it does so with clean accuracy between 86.81 and 88.20%, within the two-point budget of the gold models on every seed. So partial repair is a real mitigation. What it is not is an amplifier: the poison it fails to remove is no easier to detect than before. The pre-registered gate fails on its AUC clause on every seed; it also fails the control clause, because the controls are at chance and therefore not distinguishable from the repair in the direction the hypothesis predicts.

TABLE XII
E9: THE AMPLIFICATION EFFECT DOES NOT SURVIVE AT FULL SCALE.

(a) AUC separating missed poison from clean class-0 data

Condition	Seed 0	Seed 1	Seed 2
Base	0.494	0.499	0.496
Repair, lr 0.005	0.470	0.497	0.487
Repair, lr 0.01	0.468	0.476	0.499
Clean control, lr 0.005	0.492	0.499	0.501
Clean control, lr 0.01	0.495	0.503	0.501

(b) Repair outcomes (lr 0.005 / lr 0.01)

Seed	ASR (%)	Clean acc. (%)
0	11.1 / 33.0	88.2 / 86.8
1	15.9 / 17.7	88.1 / 86.9
2	21.3 / 29.6	88.1 / 87.3

10% poison, simulated recall-0.5 flags. AUC separates the 2,500 *missed* poison samples from clean class-0 data. Partial label repair lowers ASR but does not raise AUC above the base model by more than 0.003 in any seed; the matched clean-removal controls keep ASR at 99.94–100.00% and clean accuracy at 86.7–87.8%, and their AUC sits at chance. The pre-registered promotion gate ($\geq +0.10$ AUC, ≤ 2 points clean loss, beating control) fails on every clause.

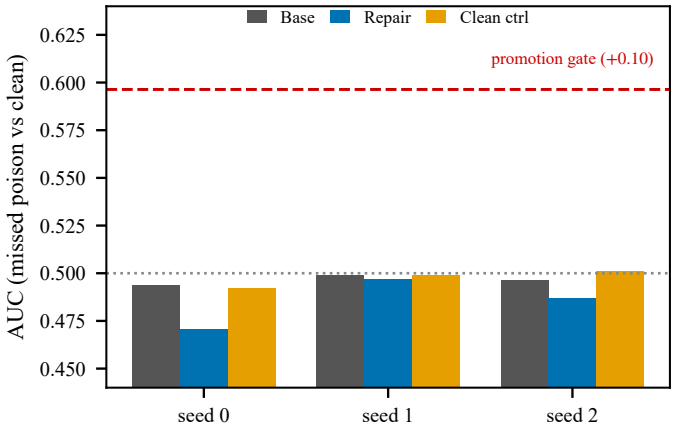


Fig. 14. AUC separating missed poison from clean target-class data at 10% poison, three seeds. Partial repair (red) and matched clean-removal controls (orange) both sit at chance; the dashed line is the pre-registered promotion gate at base +0.10.

E. What the Failures Have in Common

Both refinements try to avoid the recipe's dependence on the trigger, and both fail for the same underlying reason: they need a signal that distinguishes poison from clean data without knowing what the trigger is. CaR's proxy is a different pattern and is never suppressed; the amplification hypothesis assumes that the residual poison becomes more salient, and it does not. The only method in our suite that crosses the 5% threshold under imperfect recall is the one that uses the true trigger. That is a negative result about the state of post-training removal, and we report it as one.

IX. Durability and Efficiency-Oriented Baselines

A. Relearning the Removed Backdoor

A removal that leaves the model unable to express the backdoor today may still leave it able to relearn it tomorrow. We test this on the gold model, the oracle-deleted network that every removal method is trying to approximate. The probe gives the model one hundred freshly poisoned samples, trains for a small number of steps, and measures ASR. The samples are generated with the same trigger and target as the original attack but were not in the original training set.

TABLE XIII
E5: DURABILITY PROBES ON THE ORACLE-DELETED MODEL.

(a) Relearning: ASR (%) after k steps on 100 fresh poison samples

Setting	Step 0	Step 5	Step 10	Step 50
R18/BadNets	1.2	25.2	100.0	99.9
R18/Blend	0.6	96.0	99.3	98.7
VGG/BadNets	1.2	4.7	97.7	99.9
VGG/Blend	0.6	78.2	99.9	98.9

(b) Representational probes on the same model

Setting	MIA AUC	KL clean	KL trigger
R18/BadNets	0.495	0.000	0.0
R18/Blend	0.495	0.000	0.0
VGG/BadNets	0.491	0.000	0.0
VGG/Blend	0.493	0.000	0.0

The gold (oracle-deleted) model, i.e. the best case any removal method can reach. One hundred fresh poisoned samples restore ASR to $\geq 97.6\%$ within 10 steps in all four settings; the loss-threshold MIA sits at chance (0.49–0.50); and the clean-vs-trigger KL gap that characterises a live backdoor (0.11–0.16 vs. 12–28 in the poisoned model) has vanished. Suppression, not erasure.

Table XIII and Fig. 15 report the result. At step zero the gold model's ASR is 0.6–1.2%. After five steps it is 4.7% for VGG-16-BN/BadNets, 25.2% for ResNet-18/BadNets, 78.2% for VGG-16-BN/Blend, and 96.0% for ResNet-18/Blend. After ten steps every setting is at or above 97.6%, and it stays there through fifty steps. One hundred samples are 0.2% of the training set. The removed backdoor is not gone; it is dormant, and it returns as soon as the association is shown again.

The representational fingerprint tells the same story from the other side. A loss-threshold membership-inference attack on the gold model has AUC 0.491–0.496, indistinguishable from chance, which is what we would expect if the poisoned samples had been properly removed. The probe is deliberately simple; stronger membership-inference attacks exist [71], so the chance-level result should be read as the absence of an easy membership signal rather than as proof that none exists. The Kullback–Leibler divergence between the model's outputs on clean and trigger-stamped inputs is zero in both cases, whereas the poisoned model shows 0.110–0.157 on clean inputs and 12.0–28.2 on trigger-stamped inputs. By both standard tests the removal looks complete. The relearning probe shows that it is not.

This is the vision-domain analogue of the language-model result of Hu et al. [31], and it has a practical consequence for evaluation: a removal claim that does not include a relearning probe is a claim about the model's current behaviour, not about the data it was trained on. We did not run the relearning probe on the recipe's output, because the recipe keeps and relabels the flagged data rather than deleting it; whether its dormancy is longer or shorter than the gold model's is an open question that our records cannot answer.

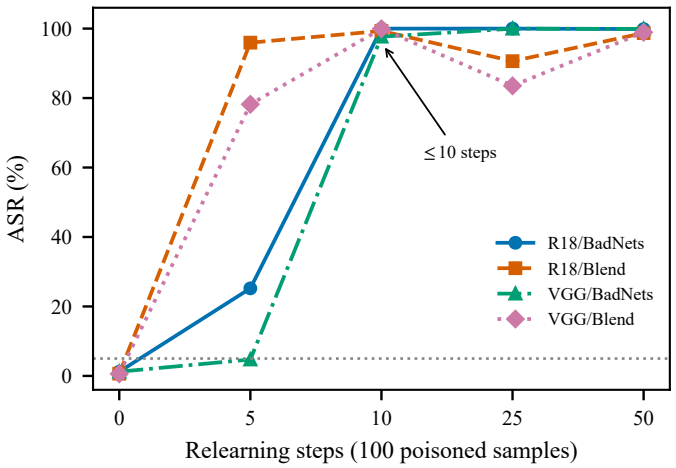


Fig. 15. ASR of the oracle-deleted gold model after k gradient steps on one hundred fresh poisoned samples. All four settings return to at least 97.6% ASR within ten steps.

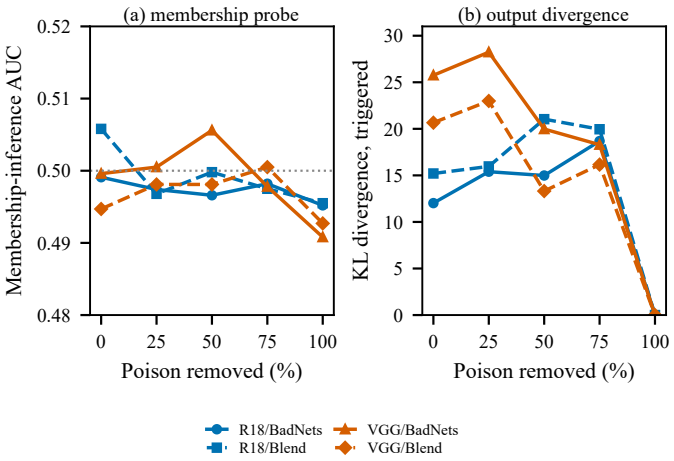


Fig. 16. Representational probes on the E2 models as a function of the fraction of poison removed. (a) Membership-inference AUC is at chance at every fraction, including the fully poisoned model. (b) KL divergence between clean and trigger-stamped outputs is high while any poison remains and zero only at full removal. Neither probe distinguishes suppression from erasure.

B. What the Probes Can and Cannot Certify

Fig. 16 places both probes on the same axis, the fraction of poison removed. Panel (a) is the membership-inference AUC. It is at chance at every fraction, including the fully poisoned model, so on these models it cannot certify anything about poison: the probe never had a signal to lose. This matters because an AUC near 0.5 is sometimes reported as evidence of successful unlearning. Here it is evidence only that the training procedure leaves no detectable membership trace, which was already true before removal. Panel (b) is the KL divergence between the model's outputs on clean and trigger-stamped inputs. It stays between 12.0 and 28.2 while any poison remains and falls to zero only when every poisoned sample is removed. Unlike the membership probe, it tracks the poison, but it certifies current behaviour rather than the absence of the capability: at 100% removal it reads zero on the same gold model that relearns to 97.7% ASR within ten steps.

C. SISA

SISA-5 is the strongest efficiency-oriented alternative in our suite: it removes the backdoor at full recall, reaching ASR 3.04% with clean accuracy 70.38%. The clean accuracy is 18.3 points below the frozen gold model, so the run fails the joint criterion by a wide margin. At recall 0.5 the ASR is 100.00% and the clean accuracy 67.49%: the method inherits the same recall cliff as deletion, on top of its sharding cost. The pilot runs with ten and twenty shards are worse still (50.25% clean at ASR 6.26%, and 37.59% at 8.04%), which is the expected direction: more shards mean less data per constituent model and a larger utility penalty. SISA's guarantee is about the data that is forgotten, not about the data that is missed, and it does not change the cliff.

D. Fine-Pruning

Fine-pruning removes the filters with the smallest average activation on clean data. On ResNet-18/BadNets, pruning 40% of channels leaves ASR at 63.64% and clean accuracy at 65.50%; pruning 60% reaches ASR 0.00% with clean accuracy 61.30%. On VGG-16-BN/BadNets the method needs 90% of channels before ASR falls to zero, at which point clean accuracy is 67.85%. The base models in this block are weaker than the frozen gold (69.5% and 71.0% clean), because the block was run under a shorter schedule, so the absolute numbers are not directly comparable with Table IV. The relative cost is: 8.2 points for ResNet-18 and 3.1 points for VGG-16-BN from their own bases, plus the 18–19-point gap between those bases and the frozen protocol. Fine-pruning is also trigger-agnostic in the wrong sense: it can only remove a backdoor that is visible as dormant neurons, and it has no mechanism for poison that is distributed across the network.

E. Pilot Baselines

Table XIV also lists rows from the earlier pilot protocol. They are not pooled with the frozen results, and we mark them as pilot rows. Three of them are informative. Partial deletion of 25%, 50%, or 75% of the poison leaves ASR between 99.86% and 100.00%, which is the same step seen in E2. Gradient ascent in the pilot leaves ASR at 99.93% on ResNet-18 and 99.99% on VGG-16-BN (mean 99.96%), consistent with the normalisation artefact diagnosed in Section VIII. Label repair without inoculation averaged ASR 0.46% at 83.37% clean accuracy over two pilot runs (0.54%/85.55% on ResNet-18 and 0.38%/81.19% on VGG-16-BN), a 6.6-point cost against that protocol's approximately 90% baseline. It did not reproduce under the frozen protocol, where label repair alone gave 8.63–16.36% ASR with clean accuracy within 1.5 points. We report both, and we do not use the pilot row to support a success claim.

F. Summary of the Baselines

The baselines divide into two groups. Methods that delete data and retrain (SISA, deletion) can remove the backdoor

TABLE XIV
EFFICIENCY-ORIENTED AND SHARDED BASELINES.

Method	Recall	Clean	ASR
SISA-5 (this protocol)	1.0	70.38	3.04
SISA-5 (this protocol)	0.5	67.49	100.00
SISA-5 (pilot)	oracle	63.83	3.79
SISA-10 (pilot)	oracle	50.25	6.26
SISA-20 (pilot)	oracle	37.59	8.04
Fine-pruning R18, 40%	–	65.50	63.64
Fine-pruning R18, 60%	–	61.30	0.00
Fine-pruning VGG, 90%	–	67.85	0.00
Gradient ascent (pilot)	–	87.62	99.96
Label repair FT (pilot)	–	83.37	0.46
Partial deletion 25% (pilot)	–	89.97	99.98
Partial deletion 50% (pilot)	–	90.13	99.96
Partial deletion 75% (pilot)	–	90.08	99.96

Reviewer-facing baselines. SISA and fine-pruning rows are measured; pilot rows use the minor project's earlier protocol (clean accuracy ≈90%) and are shown for completeness, not pooled with the frozen results. Pilot values are means over the runs recorded in the master results table (two architecture variants for SISA and repair, four runs for partial deletion). SISA's clean accuracy is 18 points below the frozen gold model already at r=1; fine-pruning needs 60% (ResNet) or 90% (VGG) of channels pruned before ASR falls, by which point clean accuracy has dropped 8 and 3 points respectively.

at full recall, but they pay a utility cost that ranges from 18 points for SISA-5 to a fraction of a point for deletion, and they inherit the recall cliff. Methods that edit the model without a trigger-aware objective (fine-pruning, gradient ascent, retain fine-tuning) either need to remove most of the network or do not remove the backdoor at all. None of them reaches the joint criterion at a cost below retraining with full recall. The recipe is the only method that reaches it at recall 0.5, and it does so with trigger knowledge that the baselines do not require.

X. CROSS-DOMAIN REPLICATION: RADIO MODULATION CLASSIFICATION

A. Motivation

The recall cliff could be a property of image classifiers, of convolutional feature hierarchies, or of the particular detectors we use. To test whether it is a property of deletion under partial poison, we repeat the core experiment in a different modality: automatic modulation classification (AMC) on RadioML 2016.10a [41], [42]. AMC is a natural second domain for this question. Models are trained on large captures that may come from crowd-sourced or third-party receivers, a poisoned capture is cheap to produce, and a backdoor that maps a specific waveform to a fixed modulation label is a plausible integrity attack on a receiver. Adversarial and poisoning threats to wireless classifiers are an active concern in their own right [72], [73], but backdoor removal under imperfect detection has not, to our knowledge, been characterised as a function of recall in this domain. The defence pipeline is the same: detect, form a forget set, remove.

The protocol is summarised in Section V. Eleven modulation classes, SNR from -20 to 18 dB, 2×128 I/Q windows, unit-energy normalisation, a three-block one-dimensional CNN, 5% poison to BPSK, and a fixed burst trigger that sets I=0.9 and Q=-0.9 on the last eight samples. Three seeds vary poison placement and initialisation; the 80/20 split is fixed. The same four methods and the same (r,p) cells are used as in the image study, including the precision-0.7 probe at r=0.75.

One feature of this domain changes how the results must be read. RadioML is evaluated per SNR, and at low SNR the clean classifier is close to chance. We report per-SNR numbers because they are the domain's native metric, but we mark the near-chance region explicitly and restrict the interpretation of the recall tolerance to SNR at or above -6 dB. A defence that reports a low ASR on a classifier that cannot classify is not a defence.

B. The Attack and the Base Model

The poisoned base model reaches ASR 99.54% at 57.66% clean accuracy, and its per-SNR ASR is between 98.61% and 100.00% at every SNR from -20 to 18 dB. That flatness is expected for a trigger that overwrites the end of the window: the model can learn to read the burst regardless of the noise on the rest of the signal. Clean accuracy, by contrast, depends strongly on SNR: 8.45% at -20 dB, 10.5% at -16, 23.4% at -10, 48.2% at -6, 63.2% at -4, 82.2% at 0, and 85–87% from 4 dB upward. At -20 dB the classifier is at chance (9.1% for eleven classes), so an ASR of 100% is not evidence that the backdoor is sophisticated; it is evidence that the trigger is easier to read than the signal.

C. Deletion Replicates the Cliff

Table XV and Fig. 17 report the removal results. Deletion works when the forget set is complete: retraining on the retained set with every poisoned sample removed reaches ASR 2.90 ± 1.43% at r=1 with clean accuracy 56.83 ± 0.57%. At r=0.75 its ASR is 97.16 ± 1.84%; at r=0.5 it is 98.41 ± 2.18%. The step is in the same place as in CIFAR-10: full recall works, three-quarters recall does not. Fine-tune never works, at any recall, with ASR 73.87–99.66%. On the SNR-averaged cell, the two trigger-aware methods work down to r=0.5: the recipe reaches 0.83 ± 0.17%, 2.35 ± 0.43%, and 4.02 ± 0.86% at the three recall levels, and inoculation-only reaches 0.28 ± 0.05%, 1.60 ± 0.23%, and 3.83 ± 0.23%. At the precision-0.7 cell (r=0.75), deletion is at 96.9–99.9% ASR, the recipe at 1.5–2.2%, and inoculation-only at 1.4–1.8%.

Two differences from the CIFAR results deserve comment. First, inoculation-only succeeds in this domain, whereas on CIFAR-10 it left ASR at 17.67%. The trigger here is a simple additive burst on two channels, and the network's response to it can be overwritten by presenting the burst with the correct label; the CIFAR patch is spatial and interacts with many channels, and inoculation alone is not enough. The cross-domain study therefore replicates the deletion cliff and the recipe's tolerance, but it does not replicate the CIFAR ordering between the two trigger-aware methods. We report the difference rather than pooling the domains.

Second, the radio task has a much lower clean-accuracy ceiling than CIFAR-10, which compresses the utility side of the success criterion. A two-point budget around a 56% baseline is a tighter relative constraint than a two-point budget

TABLE XV
CROSS-DOMAIN REPLICATION ON RADIOML AMC.

Method	r=1.0	r=0.75	r=0.5	**Clean acc**
Deletion	2.90±1.43	97.16±1.84	98.41±2.18	56.75±0.79
Fine-tune	73.87±2.33	99.48±0.35	99.66±0.14	58.55±0.38
Recipe	0.83±0.17	2.35±0.43	4.02±0.86	57.36±0.50
Inoc. only	0.28±0.05	1.60±0.23	3.83±0.23	58.40±0.30

RadioML 2016.10a, 3 seeds, 5% poison, synthetic I/Q trigger. Cells are ASR (%) mean±s.d. over SNR and seeds; the last column is clean accuracy over all runs. The poisoned base reaches ASR 99.54% at 57.66% clean accuracy. The r=1.0 cell of the deletion row is the gold retrain. The recall cliff reproduces: deletion fails at $r \leq 0.75$ (per-SNR ASR 90.3–100% at r=0.75 and 89.6–100% at r=0.5), while trigger-aware repair keeps the SNR-averaged ASR below 5% to r=0.5.

around 89%, but all methods here stay within about two points of the gold clean accuracy, so the verdicts are driven by ASR. This is the regime in which the recall cliff is easiest to see.

D. The Tolerance as a Function of SNR

Fig. 18 plots γ^*, the smallest recall at which a method reaches ASR $\leq 5\%$, as a function of SNR. Unlike r^*, γ^* carries no utility budget: below -10 dB the clean task is itself near chance, so a two-point accuracy budget would not be meaningful. Clean accuracy is reported separately in Table XV. The oracle retrain has γ^*=1.0 wherever it is defined, and it is undefined between -16 and -6 dB because the oracle model itself sits above 5% ASR there (5.45–7.87%). That undefined region is not a failure of the retrain; it is a statement that the attack is weak enough at those SNRs that the 5% threshold cannot separate a defended model from an undefended one. The two endpoints at -20 and -18 dB, where the oracle reports 4.26% and 4.63%, are also degenerate: gold clean accuracy there is 8.4% and 10.3%, at or near chance. We mark the whole region from -20 to -10 dB as near-chance in the figure.

For the trigger-aware methods γ^* is defined in every bin, and the profile is U-shaped: the requirement is lowest at high SNR, rises through the mid range, and is uninformative where the clean task is near chance. The recipe has γ^*=0.75 at -12 dB, 1.0 at -10 to -4 dB, 0.75 at -2 dB, and 0.5 from 0 dB upward. Inoculation-only has a similar profile, between 0.5 and 0.75. Fine-tuning has no γ^*: its best cell never falls below 48.5%. The practical reading is narrower than a single recall number. Above 0 dB, recall 0.5 suffices for trigger-aware repair; at -12 and -2 dB the requirement is 0.75, and between -10 and -4 dB it is 1.0; below -10 dB the clean task is near chance and the criterion is vacuous. Deletion is insufficient at every SNR at any recall below 1.0. A detector requirement stated as one number is therefore itself an average over the operating SNR; the deployment quantity is γ^* as a function of SNR.

E. Per-SNR Structure

Fig. 19 shows every cell as a heatmap. Three structures are visible. First, the fine-tune row is saturated at the partial-recall columns and weakens only at r=1, where it falls from about 100% in the near-chance bins to 48.5% at 18 dB. That is the CIFAR-10 fine-tuning result at higher resolution: even its best cell is ten times the 5% threshold. Second, the retrain row, which is the oracle at r=1 and partial deletion below it, turns from blue to red between the first and second columns at every SNR: it is at 90.3–100% at r=0.75 and 89.6–100% at r=0.5. Third, the two trigger-aware rows are blue across the band, but not uniformly. The lighter patch between -12 and -2 dB at r=0.5 is the mid-SNR hump: the recipe's per-SNR ASR rises to 13.8% at -6 dB, even though the SNR-averaged cell mean is 4.02%.

Fig. 20 explains why the left edge of the heatmap cannot be read as a defence. In the shaded region below -10 dB the three curves are indistinguishable, and at -20 and -16 dB they sit within 1.5 points of chance. Above -6 dB the recipe tracks gold within 4.3 points, so the mid-SNR hump is a real removal failure on a task the network can perform, not a side-effect of chance-level features.

F. What the Radio Study Does and Does Not Show

The study shows that the recall cliff is not specific to images, to two-dimensional convolutions, or to the two CIFAR detectors. It appears in a one-dimensional I/Q classifier with a different optimiser, a different poison mechanism, and a per-SNR evaluation. It also shows that the recipe's tolerance is not an artefact of the CIFAR architecture, although the simpler trigger makes inoculation alone sufficient here.

It does not show that the result transfers to over-the-air signals. The trigger is a digital burst applied before normalisation, not a hardware impairment; the model is small; the dataset is a standard benchmark with a fixed split; and the per-SNR bins contain different numbers of samples. These are ordinary external-validity limits, and we list them rather than claim more than the experiment supports. Within those limits, the radio study is the strongest evidence in the paper that the cliff is a property of deletion under partial poison rather than a property of one benchmark.

XI. DISCUSSION

A. The Cliff Is About Residual Poison

The grid and the partial-deletion block agree on a simple statement: what matters is whether any poison remains, not how much was removed. A model trained on 2,250 of 2,500 poison samples behaves like a fully poisoned model; a model trained on 125 behaves like a nearly clean one. The transition is sharp enough that the difference between a detector with 90% recall and one with 100% recall is the difference between no defence and a complete defence. This is why the success tolerance of deletion is exactly r^*=1.00 rather than a value slightly below one.

The result is an empirical statement about this family of attacks and models, not a theorem. A trigger with a smaller effective support, a lower poison rate, or a model with more capacity might move the step. But the step was in the same place across two architectures, two triggers, and two modalities, and it was visible from both directions (increasing recall in the grid, increasing residual poison in E2). We would be surprised if it moved far under moderate changes to the protocol.

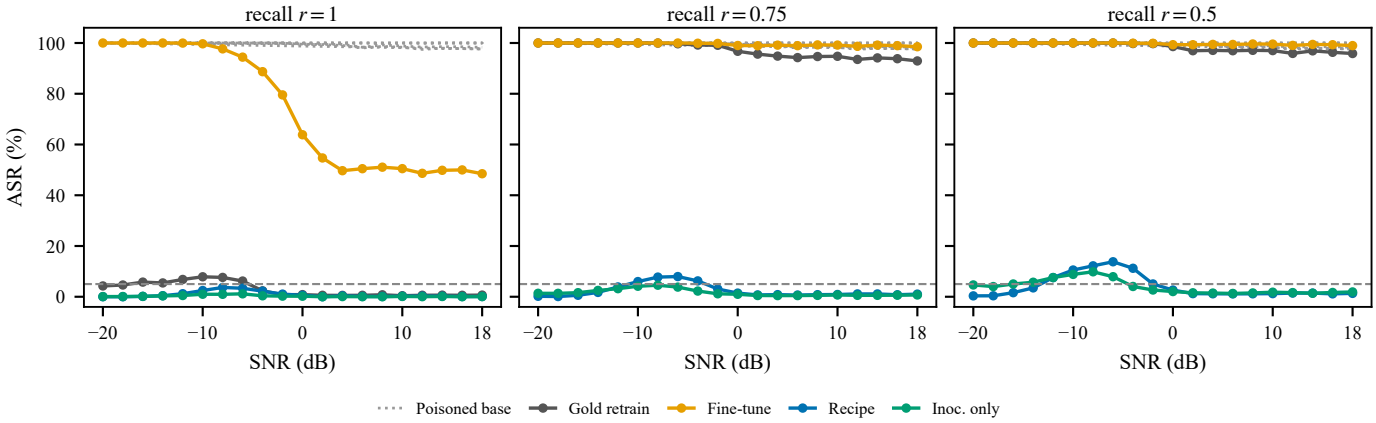


Fig. 17. ASR versus SNR on RadioML 2016.10a for recall 1.0, 0.75, and 0.5 (mean over three seeds). Deletion succeeds only at full recall (ASR 2.9%); at partial recall its ASR is 90.3–100%. Fine-tuning never removes the backdoor; both trigger-aware methods stay at or below 5% at high SNR, with the recipe rising at mid SNR as Fig. 19 shows.

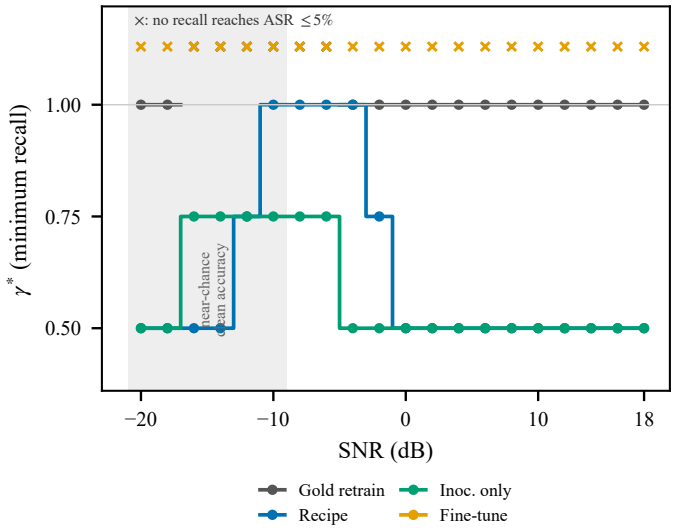


Fig. 18. γ^* (minimum recall reaching ASR $\leq$ 5%) versus SNR. Crosses mark SNRs where no recall qualifies. The shaded region is near-chance clean accuracy, where the 5% criterion is vacuous.

B. Detector Engineering Should Optimise Recall

The precision axis is nearly flat. At fixed recall, adding clean false positives to the forget set changes ASR by less than a point for deletion and by a small, non-monotone amount for the recipe. The cost appears on the utility side, where the recipe loses at most about 0.6 points across the precision cells. This suggests a division of labour: detector design should maximise recall subject to a precision floor that keeps the forget set affordable, and removal design should assume that the forget set will contain clean data.

The activation-clustering result at 10% poison is a warning about detector design as much as about removal. The detector returns nothing at exactly the poison rate where the attack is strongest, because its cluster-size threshold assumes that poison is a small minority. A detector that cannot flag a large poison fraction is not conservative; it is silently permissive. Reporting recall and precision at a single poison rate would not have revealed this.

C. Why Trigger-Aware Repair Tolerates Half the Poison

The recipe does not need to identify every poisoned sample because its inoculation stage changes the trigger-to-target association for the whole network. Label repair removes the label noise that would otherwise re-teach the association on the flagged subset; inoculation teaches the network that the trigger is uninformative on a large sample of clean data. The two mechanisms cover different failures: repair handles the poison that was found, inoculation handles the poison that was missed, up to a point. The balance breaks at r=0.25, where three-quarters of the poison keeps its original label and the inoculation signal is not enough.

This also explains why the recipe's clean-accuracy cost grows with architecture depth and trigger strength. Inoculation is a data-dependent regulariser, and the network must fit both the retained data and the re-labelled trigger examples. The cost is small (about one point) on ResNet-18 and larger (up to 4.1 points) on VGG-16-BN with Blend, which is the setting with the largest distribution shift between clean and trigger-stamped inputs.

D. The Known-Trigger Bound Is the Central Limitation

The recipe's tolerance is bought with the true trigger, and the synthesis experiment shows that the purchase cannot be avoided by estimating the trigger from the model. On both CIFAR attacks the synthesised mask had zero or near-zero overlap with the true patch, and inoculation with it left ASR above 95%. This places the recipe on the ceiling side of the problem: it is a strong upper bound on what post-training repair can do, and it is not a blind defence.

The bound matters for how the paper should be cited. A reader should not take away "repair removes backdoors at 50% recall". The correct takeaway is "repair removes backdoors at 50% recall if the trigger is known; if it is not, no method in this study does". The first statement is a benchmark result. The second is the deployment result.

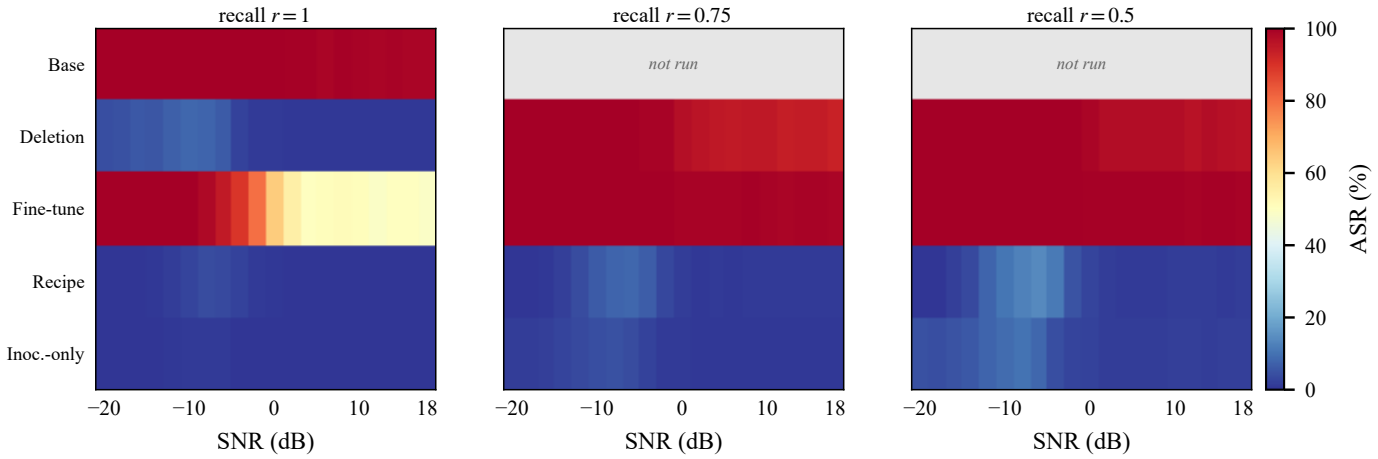


Fig. 19. ASR by SNR and recall for the radio study (mean over three seeds). The retrain row is the oracle at r=1 and partial deletion at the lower recalls; the base row was not run at partial recall (grey). Fine-tuning is saturated except at full recall above 0 dB; both trigger-aware methods are blue across the band, with a lighter patch for the recipe at mid SNR.

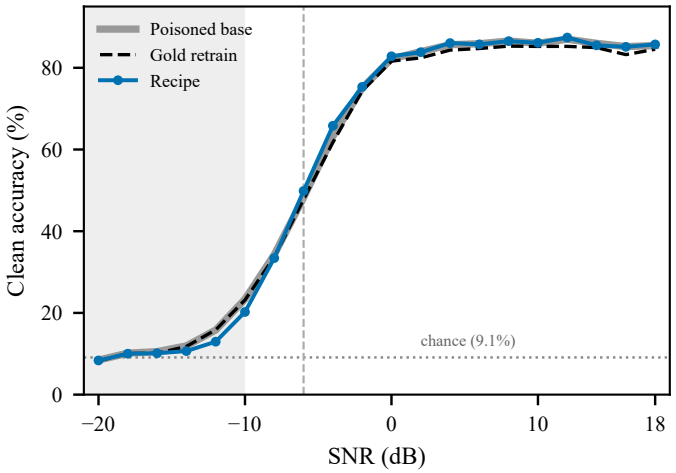


Fig. 20. Clean accuracy versus SNR for the poisoned base, the gold retrain, and the recipe at r=1. The three curves are nearly indistinguishable; the shaded region below −10 dB is near chance and the dashed vertical line marks −6 dB, the lower edge of the interpretable range.

E. Relation to Prior Work

Our findings are consistent with the negative result of Pawelczyk et al. for approximate unlearning [30], and they add a controlled curve that separates the effect of missed poison from the effect of deleted clean data. They are also consistent with the relearning result of Hu et al. [31]: the oracle-deleted model looks clean by membership inference and by output divergence, yet relearns the backdoor in ten steps. The difference is that our probe is a vision classifier and the poison is a trigger, not a text corpus, so the dormancy is not a language-model artefact.

The ascent artefact is, as far as we know, under-reported. Evaluations of unlearning that use feature-space ascent without a normalisation control may be measuring batch-norm corruption. We recommend that any ascent-based removal result be reported alongside a run with batch-norm statistics frozen to the retain set.

F. A Procedure for Detector-Bounded Removal

The results support a short procedure for anyone who has to remove a backdoor after training, and each step is tied to a measurement in this paper.

1. Measure recall on ground-truth poison, not detection F1. The quantity that predicts removal outcome is the fraction of poison in the forget set. A detector report that quotes only F1 or only a threshold hides it. A small poisoned pilot with known poison labels gives the recall curve directly and costs a fraction of the removal runs it will govern.

2. Without trigger knowledge, treat deletion as all-or-nothing. In our grid the smallest tested recall at which retraining on the retained set passes the joint criterion is 1.00; at 0.95 the ASR is low but the clean-accuracy cost already fails the criterion, and at 0.90 the backdoor is at full strength. A detector whose upper confidence bound on recall does not reach the top of that cliff should not be used to justify a deletion-based removal claim.

3. With trigger knowledge, repair is an option, with a budget. The recipe holds ASR below 5% down to r=0.5 at precision 1.0 and below 3% at precision 0.7, at a clean-accuracy cost of about 1.3 points on the ResNet-18 grid; on the transferred settings of Section VII the cost rises to 4.09 points in the hardest case (VGG-16-BN with Blend at r=0.5). The budget, not the average, is what a deployment decision needs.

4. Validate with relearning, and control for batch norm. A removal claim that stops at ASR and membership inference can be satisfied by suppression: our oracle-deleted model returns to 97.7% ASR within ten steps. Any ascent-based result should be reported next to a run with batch-norm statistics frozen, since without that control the apparent removal may be normalisation damage.

Nothing in this procedure requires a new method. It requires reporting the detector's recall, the precision at which it was measured, whether the trigger was known, and a relearning probe. Those four items are enough to make removal results comparable across papers.

G. What Would Change the Conclusion

Three developments would move the conclusion from negative to positive. A detector with recall at or near 1.0 at usable precision would make deletion viable, and the cliff would become irrelevant. A trigger-synthesis method with high IoU on realistic attacks would make the recipe blind, and its tolerance would become deployable. An amplification signal that survives matched controls and a utility budget would make iterative detect-remove loops viable. None of these was observed here. Until one is, the honest engineering advice is to invest in provenance and duplicate detection before training, because post-training removal cannot compensate for a detector that misses poison.

XII. Limitations and Threats to Validity

Benchmark coverage. The CIFAR-10 study covers two architectures (ResNet-18 and VGG-16-BN), two dirty-label attacks (BadNets and Blend), one target class, and a 5% poison rate (10% in the amplification block). It does not cover deeper or wider networks, vision transformers, ImageNet-scale data, or clean-label attacks. The cliff could move under a much lower poison rate, where the residual poison after a 0.9-recall detection is only a handful of samples, or under a trigger whose effective support is larger than a 4×4 patch. We have no evidence either way, and we do not extrapolate.

Simulated forget sets. The primary manipulation constructs the forget set directly from ground-truth poison labels with requested recall and precision. It is therefore kinder than a real detector in two ways: false positives are drawn only from non-target clean data, and the flagged poison is a uniformly random subset rather than the easiest or hardest cases. The E6 block anchors the grid to two real detector outputs and reproduces the main conclusion, but two anchors are not a survey. A detector whose errors are correlated with the poison's representation, rather than random, could produce different results; our construction cannot see that correlation.

Detector coverage. Only activation clustering and spectral signatures were swept. Neural Cleanse, ABS, STRIP, and other detectors may have different error structure. The VGG-16-BN and MobileNetV2 detector runs were degenerate and are excluded, which means the detector side of the study is, in practice, a ResNet-18/BadNets study with a cross-check on other cells. The activation clustering failure at 10% poison is a property of its thresholds; a detector with different thresholds might not fail there, although it would then have a different precision profile.

Statistical power. Three seeds per cell give coarse success counts and no confidence intervals. We report mean and standard deviation and the count of successful seeds, and we flag the one method (NegGrad+) whose distribution is bimodal. A larger number of seeds would refine the boundary between 0.90 and 0.95 recall but would not change the qualitative conclusion, which is visible in every seed. The two-point clean-accuracy budget is a modelling choice; a reader who prefers one point would move the recipe's r^* up, and a reader who prefers three points would move it down. The per-cell numbers are reported so that the verdict can be recomputed.

The r=0.95 clean-accuracy drop. At r=0.95 and p=1, deletion reaches ASR 2.84% but clean accuracy falls to 84.58%, four points below gold, with no false positives in the forget set. We did not isolate the cause. It may be an optimisation effect of retraining on a set that still contains 125 poison samples, or a seed-level interaction; the standard deviation across seeds is small (0.59), so it is not seed noise in the usual sense. We report it as measured and use the joint criterion, which fails the cell regardless of the cause.

Unrun blocks. An adaptive-attacker study (E4) was planned and not executed. Every result in this paper assumes a static attack that does not know the defence. An attacker who knows that repair and inoculation will be applied could try to construct poison that survives relabelling or that is reactivated by inoculation, and the recipe's tolerance would then be an upper bound in a stronger sense than we intend. A full hyperparameter sweep for the baselines was also not run; NegGrad+ and SCRUB were used at their default settings, and a better-tuned version might perform differently. SISA and fine-pruning were run under shorter schedules than the frozen protocol, which is why their absolute clean accuracies are lower; their relative costs are reported with that caveat.

Radio study. The trigger is a digital burst applied before normalisation, not an over-the-air impairment; the classifier is a small one-dimensional CNN; the per-SNR bins have unequal sample counts; and at SNR below -10 dB the clean model is near chance. The γ^* values in that region are reported but should not be interpreted as defence performance. The study is a replication of the cliff, not a claim about radio deployment.

Durability probe. The relearning probe used one hundred fresh poison samples, a fixed schedule, and no defence-aware optimisation. It was run on the gold model, not on the recipe's output, so it bounds the best case for erasure rather than characterising every method. A longer probe or a larger poison budget would presumably make the backdoor return faster, not slower.

Theory. We offer no guarantee. The cliff is an empirical regularity across the settings we tested, and the paper's claims are correspondingly empirical. Nothing in the results rules out a removal method, outside our suite, that provably tolerates a known fraction of missed poison under stated assumptions.

XIII. Conclusion

We asked a narrow question: once a detector has done its imperfect best, what does the removal stage actually deliver? The answer, across ten recall–precision cells, four architecture–attack combinations, three seeds, five removal families, and a second signal domain, is that it delivers what the detector's recall allows and very little more. Deletion behaves as a step function. At full recall it meets the success criterion; at 0.95 recall the attack is suppressed but the clean-accuracy cost has already broken the criterion; at 0.90 recall it leaves a model that is $\mathscr{F}$-free by construction and still 100% attackable, at every seed and every setting we tried. False positives are a second-order cost by comparison, visible only when they are numerous enough to remove a large block of clean data.

The one method that degrades gracefully is repair followed by inoculation, and the qualification matters more than the

result. It tolerates r=0.5 because its inoculation batch supplies the trigger, and a model trained to ignore that trigger ignores it whether or not every poisoned sample was found. When we replaced the known trigger with a synthesised one, the recovered patch had zero pixel overlap with the true trigger and attack success stayed above 95%. The recipe is therefore a ceiling on what trigger-aware repair can achieve against a fixed attack, not a blind post-training defence. Calibrate-then-remove inherits the same dependency and, in our sweep, could not find a threshold that separated poison from clean data; amplify-and-detect did not raise the detectability of missed poison above the promotion gate; and the apparent success of feature-space ascent was an artefact of frozen batch-norm statistics.

Two further results bound the claims that removal papers can make. Durability probes show suppression rather than erasure: the oracle-deleted model resists membership inference and output-divergence tests yet returns to 97.7% attack success within ten relearning steps. And the recall cliff is not a CIFAR artefact; on RadioML automatic modulation classification the minimum recall γ^* traces the same boundary as a function of SNR, rising where the clean task itself becomes difficult.

The practical recommendation is a reporting one. A removal result quoted at a single detector operating point cannot be interpreted, because the underlying quantity is the amount of poison left behind, not the method's name or its hyperparameters. Reporting removal as a curve over (r, p), together with the tolerance r^* and an explicit statement of whether the trigger was known, would make results comparable and would expose the trigger dependence that our experiments show is decisive. Under that reporting standard, the field's post-training methods are bounded by detection, and improving the detector remains the only lever that moves the bound.

Author Contributions

Thota Venkata Sai designed the controlled forget-set protocol and the success criterion, wrote the frozen experimental core (`core_kaggle.py`) and the seed-0 driver (`person_sai.py`), and ran the seed-0 grid, the calibrate-then-remove sweep (E3), the raw-detector evaluation (E6), the BadNets trigger-synthesis run (E7b), the full-scale amplification block (E9), and the RadioML study. **Sarang N S** ran the seed-1 grid, the VGG-16-BN transfer settings (E7), and the ascent and batch-normalisation control block (E8), and wrote `person_sarang.py`. **Divyansh Tiwari** ran the seed-2 grid, the ResNet-18/Blend transfer setting, the Blend trigger-synthesis run, and the SISA baselines, and wrote `person_divyansh.py`. **Urja Mandali** ran the residual-poison block (E2), the relearning, membership-inference, and KL probes (E5), and the pilot partial-deletion baselines, and wrote `person_urja.py`. All authors reviewed the protocol, checked the run logs against the tables, and approved the manuscript.

Artifact Map and Reproducibility

Every number in this paper is produced by a small set of files. The frozen core defines the models, the two attacks, the two detectors, the training loop, and the CSV logger. The four person drivers import that core and add only the experiment blocks assigned above; each run appends one row containing the requested and realised recall and precision, the forget-set size, clean accuracy, ASR, wall-clock time, and free-text notes. The rows are merged, validated, and read by the paper builder, which writes the LaTeX tables and the PDF figures directly. No table value is typed by hand.

The reproduction path is four commands: run the four drivers on Kaggle (2×T4, approximately 23 GPU-hours for the CIFAR blocks and one hour for the radio block), merge their CSV outputs, run the validator, then run the asset builder. The builder is deterministic given the CSVs, so the tables in this paper can be regenerated without retraining a model; `audit_numbers.py` performs the independent check and prints the value behind each claim. Model checkpoints are not distributed, because they are large and reproducible from the drivers; all result CSVs, the fixed splits, the trigger definitions, and the seed assignments are included. The repository also contains the notebook generators and the smoke tests used before the full runs, so the environment can be checked on a laptop CPU before any GPU time is spent.

References

[1] T. Gu, B. Dolan-Gavitt, and S. Garg, "BadNets: Identifying vulnerabilities in the machine learning model supply chain," *arXiv:1708.06733*, 2017.

[2] Y. Liu, S. Ma, Y. Aafer, W.-C. Lee, J. Zhai, W. Wang, and X. Zhang, "Trojaning attack on neural networks," in *Proc. NDSS*, 2018.

[3] X. Chen, C. Liu, B. Li, K. Lu, and D. Song, "Targeted backdoor attacks on deep learning systems using data poisoning," *arXiv:1712.05526*, 2017.

[4] A. Turner, D. Tsipras, and A. Madry, "Label-consistent backdoor attacks," *arXiv:1912.02771*, 2019.

[5] A. Saha, A. Subramanya, and H. Pirsiavash, "Hidden trigger backdoor attacks," in *Proc. AAAI*, 2020, pp. 11957–11965.

[6] T. A. Nguyen and A. Tran, "WaNet – imperceptible warping-based backdoor attack," in *Proc. ICLR*, 2021.

[7] K. D. Doan, Y. Lao, W. Zhao, and P. Li, "LIRA: Learnable, imperceptible and robust backdoor attacks," in *Proc. IEEE/CVF ICCV*, 2021, pp. 11966–11976.

[8] T. A. Nguyen and A. T. Tran, "Input-aware dynamic backdoor attack," in *Proc. NeurIPS*, 2020, pp. 3454–3464.

[9] E. Wenger, J. Passananti, Y. Yao, H. Zheng, and B. Y. Zhao, "Backdoor attacks on facial recognition in the physical world," *arXiv:2006.14580*, 2020.

[10] E. Bagdasaryan, A. Veit, Y. Hua, D. Estrin, and V. Shmatikov, "How to backdoor federated learning," in *Proc. AISTATS*, 2020, pp. 2938–2948.

[11] A. N. Bhagoji, S. Chakraborty, P. Mittal, and S. Calo, "Analyzing federated learning through an adversarial lens," in *Proc. ICML*, 2019, pp. 634–644.

[12] Y. Li, Y. Jiang, Z. Li, and S.-T. Xia, "Backdoor learning: A survey," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 1, pp. 5–22, 2024.

[13] M. Goldblum, D. Tsipras, C. Xie, X. Chen, A. Schwarzschild, D. Song, A. Madry, B. Li, and T. Goldstein, "Dataset security for machine learning: Data poisoning, backdoor attacks, and defenses," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 2, pp. 1563–1580, 2023.

[14] B. Wang, Y. Yao, S. Shan, H. Li, B. Viswanath, H. Zheng, and B. Y. Zhao, "Neural Cleanse: Identifying and mitigating backdoor attacks in neural networks," in *Proc. IEEE Symp. Security and Privacy*, 2019, pp. 707–723.

[15] Y. Liu, W.-C. Lee, G. Tao, S. Ma, Y. Aafer, and X. Zhang, "ABS: Scanning neural networks for back-doors by artificial brain stimulation," in *Proc. ACM CCS*, 2019, pp. 1265–1282.

[16] Y. Gao, C. Xu, D. Wang, S. Chen, D. C. Ranasinghe, and S. Nepal, "STRIP: A defence against trojan attacks on deep neural networks," in *Proc. ACSAC*, 2019, pp. 113–125.

TABLE XVI
ARTIFACT MAP. EVERY FILE LISTED IS INCLUDED IN THE ACCOMPANYING REPOSITORY; ITS README MAPS EACH NAME TO THE EXACT PATH.

Stage	File(s)	What it produces
Frozen core	`core_kaggle.py`	Model definitions (ResNet-18, VGG-16-BN, 1-D CNN), BadNets and Blend injection, activation clustering and spectral signatures, training and evaluation, CSV logging.
Drivers	`person_sai.py`, `person_sarang.py`, `person_divyansh.py`, `person_urja.py`	E1 (one seed each), E2, E3, E5–E9, SISA, pilot baselines; one CSV row per run.
Run logs	`runs_sai.csv`, `runs_sarang.csv`, `runs_sarang_v2.csv`, `runs_divyansh.csv`	Raw CIFAR runs, including degenerate detector outputs kept for the record.
Auxiliary logs	`metrics_table_urja.csv`, `relearning_summary_urja.csv`, `detection_sai.csv`, `fine_pruning_sai.csv`, `spike_repair_seed0.csv`, `spike_repair_seed1.csv`	E2 probes, E5 relearning, detector operating points, fine-pruning, amplification spike.
Radio logs	`runs_rf_study.csv`, `rf_study_per_snr.csv`, `gamma_star.csv`	RadioML run-level ASR, per-SNR cells, and γ^*.
Merge and checks	`merge_runs.py`, `validate_runs.py`, `smoke_core.py`, `smoke_rf_study.py`	Master table, schema and range validation, offline smoke tests.
Paper build	`build_paper_assets.py`, `build_extra_figures.py`, `audit_numbers.py`	All tables and figures; independent recomputation of every headline number from the CSVs.
Notebooks	`make_sai_notebook.py`, `make_sarang_fix_notebook.py`, `make_divyansh_fix_notebook.py`, `make_team_notebooks.py`, `make_RF_study_notebook.py`	Kaggle notebooks that wrap the frozen core and a person driver; the notebooks are generated, not edited by hand.

[17] B. Chen, W. Carvalho, N. Baracaldo, H. Ludwig, B. Edwards, T. Lee, I. Molloy, and B. Srivastava, "Detecting backdoor attacks on deep neural networks by activation clustering," in *Proc. AAAI Workshop on Artificial Intelligence Safety (SafeAI)*, 2019.

[18] B. Tran, J. Li, and A. Madry, "Spectral signatures in backdoor attacks," in *Proc. NeurIPS*, 2018, pp. 8000–8010.

[19] K. Liu, B. Dolan-Gavitt, and S. Garg, "Fine-pruning: Defending against backdooring attacks on deep neural networks," in *Proc. RAID*, 2018, pp. 273–294.

[20] I. Sabolić, M. Grcić, and S. Šegvić, "Seal your backdoor with variational defense," *arXiv:2503.08829*, 2025.

[21] L. Bourtoule, V. Chandrasekaran, C. A. Choquette-Choo, H. Jia, A. Travers, B. Zhang, D. Lie, and N. Papernot, "Machine unlearning," in *Proc. IEEE Symp. Security and Privacy*, 2021, pp. 141–159.

[22] L. Graves, V. Nagisetty, and V. Ganesh, "Amnesiac machine learning," in *Proc. AAAI*, 2021, pp. 11516–11524.

[23] A. Golatkar, A. Achille, and S. Soatto, "Eternal sunshine of the spotless net: Selective forgetting in deep networks," in *Proc. IEEE/CVF CVPR*, 2020, pp. 9304–9312.

[24] A. Ginart, M. Y. Guan, and G. Valiant, "Certified data removal from machine learning models," in *Proc. ICML*, 2019, pp. 3832–3842.

[25] A. K. Tarun, V. S. Chundawat, M. Mandal, and M. Kankanhalli, "Fast yet effective machine unlearning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 8231–8240, 2023.

[26] P. W. Koh and P. Liang, "Understanding black-box predictions via influence functions," in *Proc. ICML*, 2017, pp. 1885–1894.

[27] G. Pruthi, F. Liu, S. Kale, and M. Sundararajan, "Estimating training data influence by tracing gradient descent," in *Proc. NeurIPS*, 2020, pp. 19920–19930.

[28] V. M. Suriyakumar and A. C. Wilson, "Algorithms that approximate data removal: New results and limitations," in *Proc. NeurIPS*, 2022.

[29] Q. Zhao and C. Wressnegger, "Two sides of the same coin: Learning the backdoor to remove the backdoor," in *Proc. AAAI*, 2025.

[30] M. Pawelczyk, J. Z. Di, Y. Lu, G. Kamath, A. Sekhari, and S. Neel, "Machine unlearning fails to remove data poisoning attacks," in *Proc. ICLR*, 2025.

[31] S. Hu, Y. Fu, Z. S. Wu, and V. Smith, "Unlearning or obfuscating? Jogging the memory of unlearned LLMs via benign relearning," in *Proc. ICLR*, 2025.

[32] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. New York, NY, USA: Springer, 2005.

[33] A. N. Angelopoulos and S. Bates, "Conformal prediction: A gentle introduction," *Foundations and Trends in Machine Learning*, vol. 16, no. 4, pp. 494–591, 2023.

[34] R. Shokri, M. Stronati, C. Song, and V. Shmatikov, "Membership inference attacks against machine learning models," in *Proc. IEEE Symp. Security and Privacy*, 2017, pp. 3–18.

[35] A. Krizhevsky, "Learning multiple layers of features from tiny images," Dept. Comput. Sci., Univ. Toronto, Toronto, ON, Canada, Tech. Rep., 2009.

[36] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE CVPR*, 2016, pp. 770–778.

[37] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. ICLR*, 2015.

[38] L. N. Smith, "A disciplined approach to neural network hyper-parameters: Part 1 – learning rate, batch size, momentum, and weight decay," *arXiv:1803.09820*, 2018.

[39] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. ICML*, 2015, pp. 448–456.

[40] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. NeurIPS*, 2019, pp. 8024–8035.

[41] T. J. O'Shea, J. Corgan, and T. C. Clancy, "Convolutional radio modulation recognition networks," in *Proc. EANN*, 2016, pp. 213–226.

[42] T. J. O'Shea, T. Roy, and T. C. Clancy, "Over-the-air deep learning based radio signal classification," *IEEE J. Sel. Topics Signal Process.*, vol. 12, no. 1, pp. 168–179, 2018.

[43] A. Shafahi, W. R. Huang, M. Najibi, O. Suciu, C. Studer, T. Dumitras, and T. Goldstein, "Poison frogs! Targeted clean-label poisoning attacks on neural networks," in *Proc. NeurIPS*, 2018, pp. 6103–6113.

[44] P. W. Koh, J. Steinhardt, and P. Liang, "Stronger data poisoning attacks break data sanitization defenses," *Machine Learning*, vol. 111, no. 1, pp. 1–47, 2022.

[45] M. Jagielski, G. Severi, N. P. Harger, and A. Oprea, "Subpopulation data poisoning attacks," in *Proc. ACM CCS*, 2021, pp. 3104–3122.

[46] Y. Liu, X. Ma, J. Bailey, and F. Lu, "Reflection backdoor: A natural backdoor attack on deep neural networks," in *Proc. ECCV*, 2020, pp. 182–199.

[47] A. Salem, R. Wen, M. Backes, S. Ma, and Y. Zhang, "Dynamic backdoor attacks against machine learning models," in *Proc. IEEE EuroS&P*, 2022, pp. 703–718.

[48] Y. Zeng, W. Park, Z. M. Mao, and R. Jia, "Rethinking the backdoor attacks' triggers: A frequency perspective," in *Proc. IEEE/CVF ICCV*, 2021, pp. 16473–16481.

[49] Y. Yao, H. Li, H. Zheng, and B. Y. Zhao, "Latent backdoor attacks on deep neural networks," in *Proc. ACM CCS*, 2019, pp. 2041–2055.

[50] H. Wang, Z. Xiang, D. J. Miller, and G. Kesidis, "MM-BD: Post-training detection of backdoor attacks with arbitrary backdoor pattern types using

a maximum margin statistic," in *Proc. IEEE Symp. Security and Privacy*, 2024, pp. 1994–2012.

[51] B. Wu, H. Chen, M. Zhang, Z. Zhu, S. Wei, D. Yuan, and C. Shen, "BackdoorBench: A comprehensive benchmark of backdoor learning," in *Proc. NeurIPS Datasets and Benchmarks*, 2022.

[52] N. Carlini and A. Terzis, "Poisoning and backdooring contrastive learning," in *Proc. ICLR*, 2022.

[53] A. Wan, E. Wallace, S. Shen, and D. Klein, "Poisoning language models during instruction tuning," in *Proc. ICML*, 2023, pp. 35413–35425.

[54] E. Hubinger et al., "Sleeper agents: Training deceptive LLMs that persist through safety training," *arXiv:2401.05566*, 2024.

[55] T. T. Nguyen, T. T. Huynh, P. L. Nguyen, A. W.-C. Liew, H. Yin, and Q. V. H. Nguyen, "A survey of machine unlearning," *arXiv:2209.02299*, 2022.

[56] P. Maini, Z. Feng, A. Schwarzschild, Z. C. Lipton, and J. Z. Kolter, "TOFU: A task of fictitious unlearning for LLMs," in *Proc. COLM*, 2024.

[57] N. Li et al., "The WMDP benchmark: Measuring and reducing malicious use with unlearning," in *Proc. ICML*, 2024.

[58] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster, "Conformal risk control," in *Proc. ICLR*, 2024.

[59] S. Liu et al., "Rethinking machine unlearning for large language models," *arXiv:2402.08787*, 2024.

[60] B. Biggio, B. Nelson, and P. Laskov, "Poisoning attacks against support vector machines," in *Proc. ICML*, 2012.

[61] J. Steinhardt, P. W. Koh, and P. Liang, "Certified defenses for data poisoning attacks," in *Proc. NeurIPS*, 2017, pp. 3517–3529.

[62] M. Jagielski, A. Oprea, B. Biggio, C. Liu, C. Nita-Rotaru, and B. Li, "Manipulating machine learning: Poisoning attacks and countermeasures for regression learning," in *Proc. IEEE Symp. Security and Privacy*, 2018, pp. 19–35.

[63] J. Geiping, L. Fowl, W. R. Huang, W. Czaja, G. Taylor, M. Moeller, and T. Goldstein, "Witches' brew: Industrial scale data poisoning via gradient matching," in *Proc. ICLR*, 2021.

[64] C. Xie, K. Huang, P.-Y. Chen, and B. Li, "DBA: Distributed backdoor attacks against federated learning," in *Proc. ICLR*, 2020.

[65] Y. Li, X. Lyu, N. Koren, L. Lyu, B. Li, and X. Ma, "Anti-backdoor learning: Training clean models on poisoned data," in *Proc. NeurIPS*, 2021.

[66] K. Huang, Y. Li, B. Wu, Z. Qin, and K. Ren, "Backdoor defense via decoupling the training process," in *Proc. ICLR*, 2022.

[67] D. Wu and Y. Wang, "Adversarial neuron pruning purifies backdoored deep models," in *Proc. NeurIPS*, 2021.

[68] Y. Li, X. Lyu, N. Koren, L. Lyu, B. Li, and X. Ma, "Neural attention distillation: Erasing backdoor triggers from deep neural networks," in *Proc. ICLR*, 2021.

[69] W. Guo, L. Wang, X. Xing, M. Du, and D. Song, "TABOR: A highly accurate approach to inspecting and restoring Trojan backdoors in AI systems," *arXiv:1908.01763*, 2019.

[70] M. Toneva, A. Sordoni, R. Tachet des Combes, A. Trischler, Y. Bengio, and G. J. Gordon, "An empirical study of example forgetting during deep neural network learning," in *Proc. ICLR*, 2019.

[71] N. Carlini, S. Chien, M. Nasr, S. Song, A. Terzis, and F. Tramèr, "Membership inference attacks from first principles," in *Proc. IEEE Symp. Security and Privacy*, 2022, pp. 1897–1914.

[72] M. Sadeghi and E. G. Larsson, "Adversarial attacks on deep-learning based radio signal classification," *IEEE Wireless Commun. Lett.*, vol. 8, no. 1, pp. 213–216, 2019.

[73] B. Flowers, R. M. Buehrer, and W. C. Headley, "Evaluating adversarial evasion attacks in the context of wireless communications," *IEEE Trans. Inf. Forensics Security*, vol. 15, pp. 1102–1113, 2020.

Thota Venkata Sai is an undergraduate student in the Department of Electronics and Communication Engineering at Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat, India. Their research interests are in trustworthy machine learning, with a focus on data poisoning, backdoor detection, and post-training removal. They designed the controlled forget-set protocol, wrote the shared experimental driver used for the grid, and ran the seed-0 grid, the calibration, detector, trigger-synthesis, amplification, and radio studies reported in this paper.

Sarang N S is an undergraduate student in the Department of Electronics and Communication Engineering at SVNIT, Surat, India. Their interests include deep learning systems, computer vision, and the empirical evaluation of model security. They ran the seed-1 replication of the grid, the VGG-16-BN transfer settings, and the feature-space-ascent and batch-normalisation control experiments.

Divyansh Tiwari is an undergraduate student in the Department of Electronics and Communication Engineering at SVNIT, Surat, India. Their interests include machine learning for wireless systems and the robustness of learned classifiers. They ran the seed-2 replication of the grid, the ResNet-18/Blend transfer setting, the Blend trigger-synthesis run, and the SISA shard baselines.

Urja Mandali is an undergraduate student in the Department of Electronics and Communication Engineering at SVNIT, Surat, India. Their interests are in data-centric machine learning and privacy. They built the residual-poison block, the relearning, membership-inference, and KL durability probes, and the pilot partial-deletion baselines.